



LINEAR ALGEBRA AND ITS APPLICATIONS

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LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION

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PREREQUISITES:

Course Content: Prerequisites

What?

Why?



LINEAR ALGEBRA AND ITS APPLICATIONS

PREREQUISITES:



- ❖ **What** is Linear Algebra?
- ❖ Linear Algebra is the study of systems of Linear equations.
- ❖ It is concerned with vector spaces and linear mappings between such spaces.
- ❖ It includes the study of lines, planes and subspaces and is also concerned with properties common to all vector spaces.
- ❖ Linear Algebra is the geometry of n -dimensional space and its linear transformations. Thus Linear Algebra helps us to develop our geometric instinct to visualize the concepts in higher dimensions.

LINEAR ALGEBRA AND ITS APPLICATIONS

PREREQUISITES:

Why do we need to study Linear Algebra?

- ❖ Linear Algebra is used in the everyday world to solve problems in Mathematics, Physics, Biology, Chemistry, Engineering, Statistics, Economics, Finance, Psychology, Sociology, etc.,
- ❖ Applications that use Linear Algebra include the transmission of information, the development of special effects in film and video, recording of sound, Web search engines on the Internet and economic analyses.
- ❖ Linear Algebra is a study of Linear Transformations. It is used in Graph Theory, Networks, Signal Processing, Probability Theory, Real Analysis, Communication.
- ❖ Linear Algebra is used in coding theory, Artificial Intelligence, Machine Learning, Image Processing, Computer Graphics, Numerical Analysis, Control Systems, Networking, Ordinary/Partial Differential Equations.

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INTRODUCTION:



Course Content of Linear Algebra:

- ❖ **Module 1:** Matrices and Gaussian Elimination
- ❖ **Module 2:** Vector Spaces
- ❖ **Module 3:** Linear Transformations and Orthogonality
- ❖ **Module 4:** Orthogonalization , Eigen Values and Eigen Vectors
- ❖ **Module 5:** Singular Value Decomposition

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MATRICES AND GAUSSIAN ELIMINATION:

Course Content of Module 1:

- ❖ Introduction
- ❖ The Geometry of Linear Equations
- ❖ Gaussian Elimination
- ❖ Singular Cases
- ❖ Elimination Matrices
- ❖ Triangular Factors and Row Exchanges
- ❖ Inverses and Transposes
- ❖ Inverse by Gauss Jordan Method



Course Content: Introduction

What is a linear Equation?

❖ A linear equation in n variables is of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$

where $x_1, x_2, \cdots, x_n$ are **unknowns or variables**, $a_i \in \mathbb{R}$ ($i = 1, 2, \cdots, n$)

are known as the **Coefficients** of the variables x_i ($i = 1, 2, \cdots, n$)

and $b \in \mathbb{R}$ are constants.

Examples of **Linear equations** are: $2x_1 - 3x_2 = 7$; $(\sqrt{2})x_1 - (\sqrt{5})x_2 = 1$

Examples of **Non-Linear equations** are: $x_1x_2 - 3x_3 = 2$; $y = \log x + \sin x$
 $2x + 3y^{1/3} = 5$

INTRODUCTION:

What is a system of linear Equations?

❖ A **system of linear equation** is a set of linear equations involving the same **unknowns or variables** .

❖ A system of **two** equations with **two** unknowns is of the form

$$\begin{aligned}a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2\end{aligned}$$

❖ A system of **three** equations with **three** unknowns is of the form

$$\begin{aligned}a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3\end{aligned}$$

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INTRODUCTION:

Matrix Notation:

❖ A system of m equations with n unknowns is given by

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m\end{aligned}$$

This can be represented in matrix form as $Ax=b$ where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}_{m \times n} ; x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}_{n \times 1} ; b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}_{m \times 1}$$

Here A is a matrix of order $m \times n$ known as **The Co-efficient Matrix**,

$x_{n \times 1}$ is a $n \times 1$ column matrix(or vector) known as **The Vector of unknowns or variables**

$b_{m \times 1}$ is a column vector of order $m \times 1$ known as **The Constant Vector**

INTRODUCTION:

- ❖ a_{ij} is the component in the i^{th} row and j^{th} column of A.
- ❖ If $m=n$, A will be a **Square matrix of order n** and the system will have n equations with n variables.
- ❖ If all b_i 's are zero, the system is known as **Homogeneous system of Equations**.
If at least one of b_i 's is not zero then the system is said to be **Non-Homogeneous system of Equations**.

❖ Matrix $[A : b]$ =
$$\begin{pmatrix} a_{11} & a_{12} \cdots & a_{1n} : b_1 \\ a_{21} & a_{22} \cdots & a_{2n} : b_2 \\ \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} \cdots & a_{mn} : b_m \end{pmatrix}_{m \times (n+1)}$$
 Is called the **Augmented Matrix**.

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INTRODUCTION:

- ❖ **Solution to the System of Equations:** A set of values or numbers assigned for variables $x_1, x_2, \dots, x_n$ which satisfy all the equations simultaneously is defined as the **Solution** to the system of equations.
- ❖ The set of all possible solutions is called the **Solution Set or General Solution** of Linear system of equations.
- ❖ **Consistency:** A linear system of equations is said to be **Consistent** if it has a solution (unique solution or infinitely many solutions).
- ❖ A linear system of equations is said to be **Inconsistent** if it has no solution.



INTRODUCTION:

❖ **Elementary Row Operations:** In order to solve the system of linear equations using Matrix notation we define three basic operations called **Elementary Row Operations** or **Elementary Row Transformations** which are as follows:

- Multiply the entries of a row by a non-zero scalar k i.e $R_i \rightarrow kR_i (k \neq 0)$
- Replace one row by sum of itself and a non-zero scalar multiple k of another row i.e $R_i \rightarrow R_i + kR_j (k \neq 0)$.
- Interchange of any two rows i.e $R_i \leftrightarrow R_j$.

❖ **Elementary Matrix:** A square matrix of order n is called an **Elementary Matrix** E_n if it can be obtained from an Identity matrix I_n using a single row operation. i.e $I_n \rightarrow E_n$

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = E_2$$

INTRODUCTION:

❖ **Equivalent Matrices:** If two matrices A and B are such that each of them can be obtained from the other by a definite number of Elementary transformations then they are said to be **Equivalent Matrices** represented by $A \sim B$.

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & -3 & 4 \end{pmatrix} \xrightarrow{R_2 + \left(\frac{1}{2}\right)R_1} \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & 1 \\ 0 & -3 & 4 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & 1 \\ 0 & 0 & 6 \end{pmatrix} \sim B \therefore A \sim B$$

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INTRODUCTION:



- ❖ **Echelon Form Of A Matrix:** A rectangular matrix A of order $m \times n$ is said to be in **Echelon Form U** if it satisfies the following conditions:
 - The first non-zero element in every row is known as the **pivot**.
 - Below each pivot is a column of **zeros** obtained by Elementary row operations.
 - Each pivot lies to the right of the pivot in the row above. This produces a **Staircase pattern** as shown below.
 - Zero rows(if exist) appear at the bottom of the Matrix.

$$A \rightarrow \begin{pmatrix} a & b & c & d & e \\ 0 & 0 & f & g & h \\ 0 & 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{4 \times 5} = U$$

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INTRODUCTION:



- ❖ **Row Reduced Echelon Form Of A Matrix(RREF):** A rectangular matrix A in Echelon Form must further undergo the following operations to reduce to RREF :
- Every row in Echelon form must be divided by its pivot so that the first non-zero entry in every row is 1.
 - Using the pivot rows produce zeros above the first non-zero entry (i.e 1).

This RREF is denoted by R.

$$A \rightarrow U = \begin{pmatrix} a & b & c & d & e \\ 0 & 0 & f & g & h \\ 0 & 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & b/a & 0 & 0 & e/a \\ 0 & 0 & 1 & 0 & h/f \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = R$$

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INTRODUCTION:

Example:

$$A \rightarrow \begin{pmatrix} 1 & 1 & -2 & 3: & 4 \\ 2 & 3 & 3 & -1: & 3 \\ 5 & 7 & 4 & -1: & 5 \end{pmatrix}$$

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References/Links:

<https://en.wikipedia.org/wiki>





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INTRODUCTION:

Example:

$$A \rightarrow \begin{pmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 2 & 3 & 3 & -1 & : & 3 \\ 5 & 7 & 4 & -1 & : & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 0 & 1 & 7 & -7 & : & -5 \\ 0 & 0 & 0 & -2 & : & -5 \end{pmatrix} = U \rightarrow \begin{pmatrix} 1 & 0 & -9 & 0 & : & -16 \\ 0 & 1 & 7 & 0 & : & -25/2 \\ 0 & 0 & 0 & 1 & : & 5/2 \end{pmatrix} = R$$



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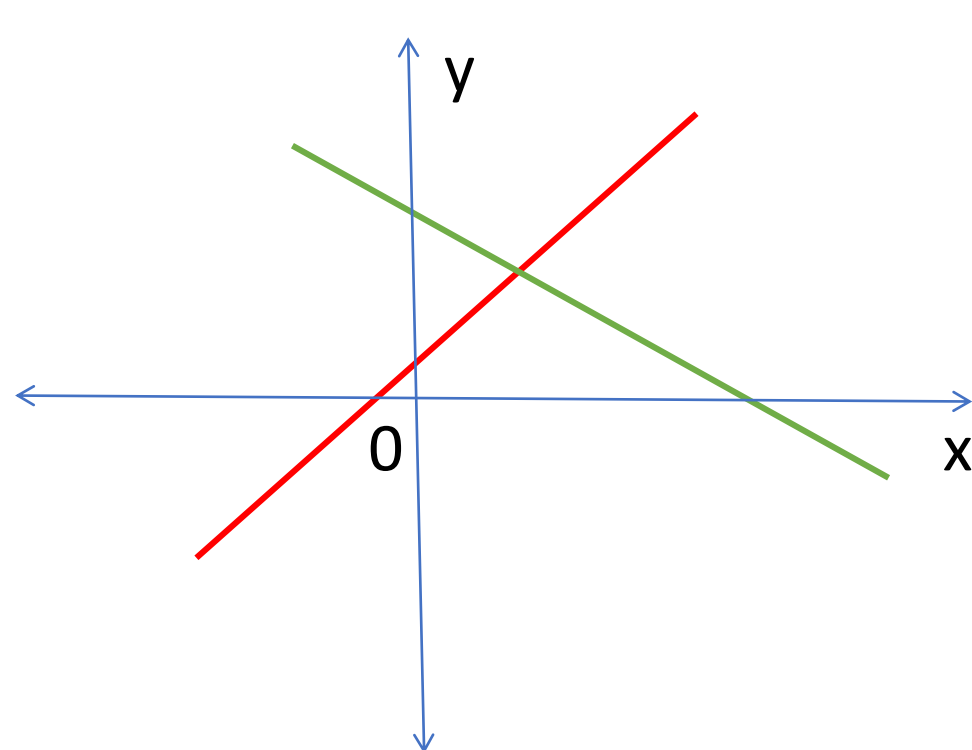
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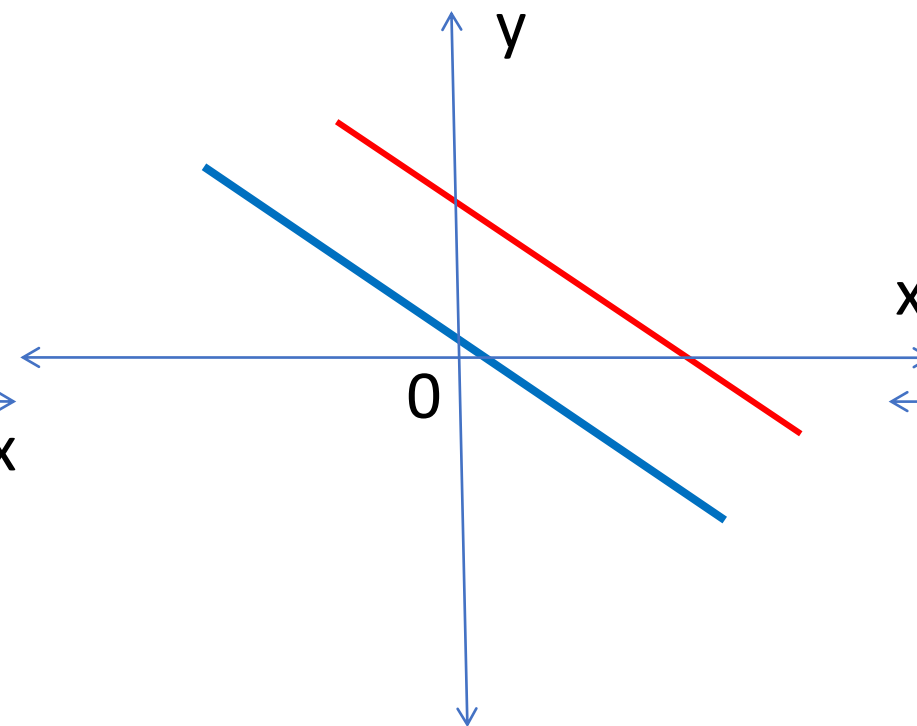
THE GEOMETRY OF LINEAR EQUATIONS:

Course Content: The Geometry of Linear Equations

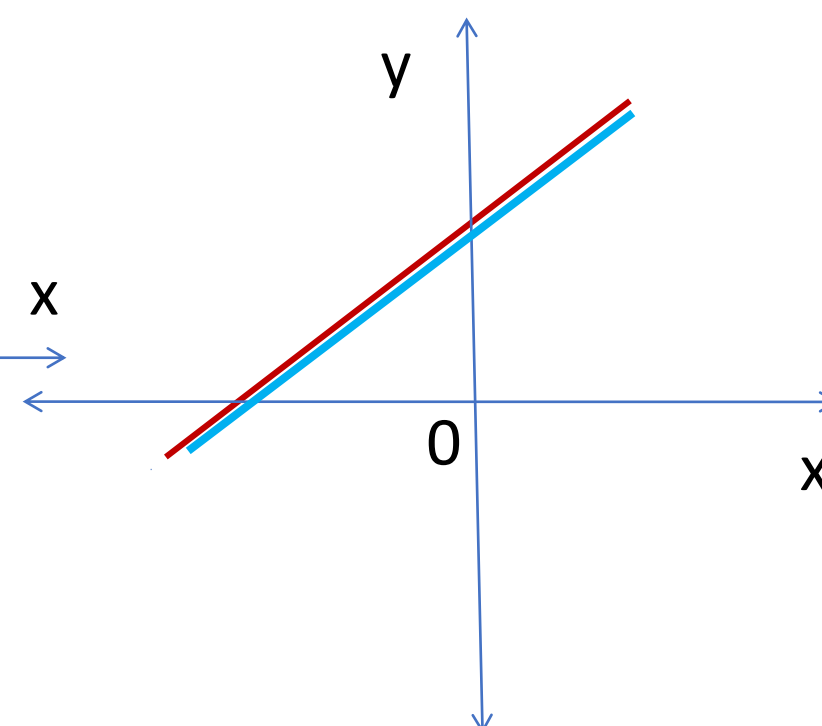
❖ SYSTEM OF 2 EQUATIONS WITH 2 VARIABLES: $a_1x + b_1y = c_1$
 $a_2x + b_2y = c_2$



Intersecting lines
Unique solution



Parallel lines
No Solution



Overlapping lines
Infinite no. of solutions

System of m linear equations with n unknowns have either a unique solution or infinite solutions or no solution.

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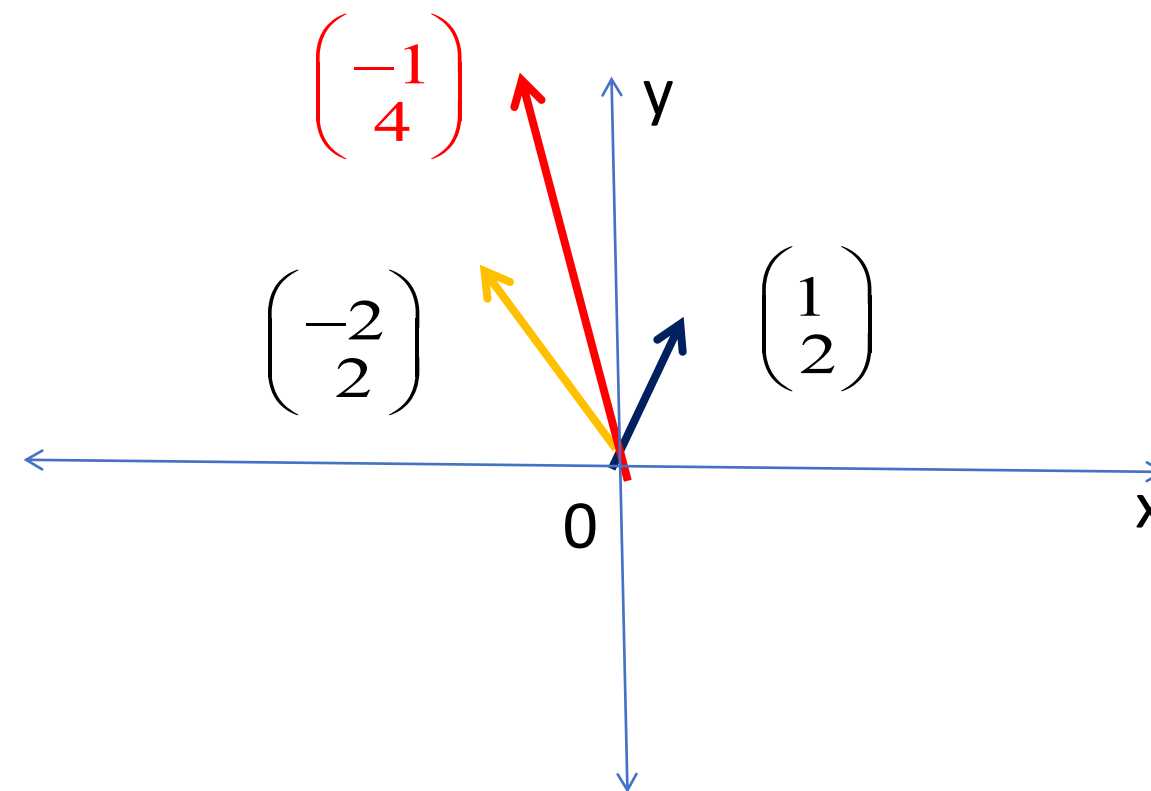
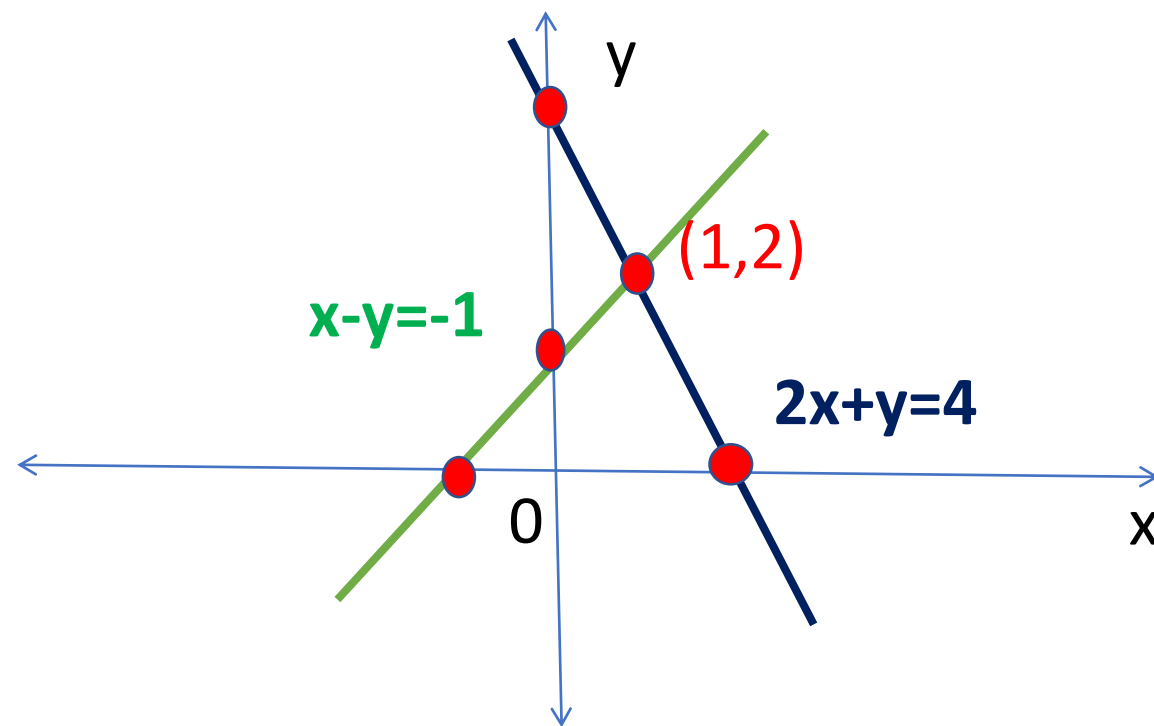
THE GEOMETRY OF LINEAR EQUATIONS:

- (i) Consider a system of 2 equations with 2 variables
- $$\begin{aligned}x - y &= -1 \\ 2x + y &= 4\end{aligned}$$
- **Row Picture:** Solving these 2 equations we get (1,2) as the point of intersection of the 2 lines. Hence this system has **a unique solution**.

❖ **Column Picture:** $x \begin{pmatrix} 1 \\ 2 \end{pmatrix} + y \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix} \Rightarrow 1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$

x=1 and y=2 will satisfy this equation.

Hence the linear combination of the column vectors on LHS produces the vector on RHS.



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THE GEOMETRY OF LINEAR EQUATIONS:

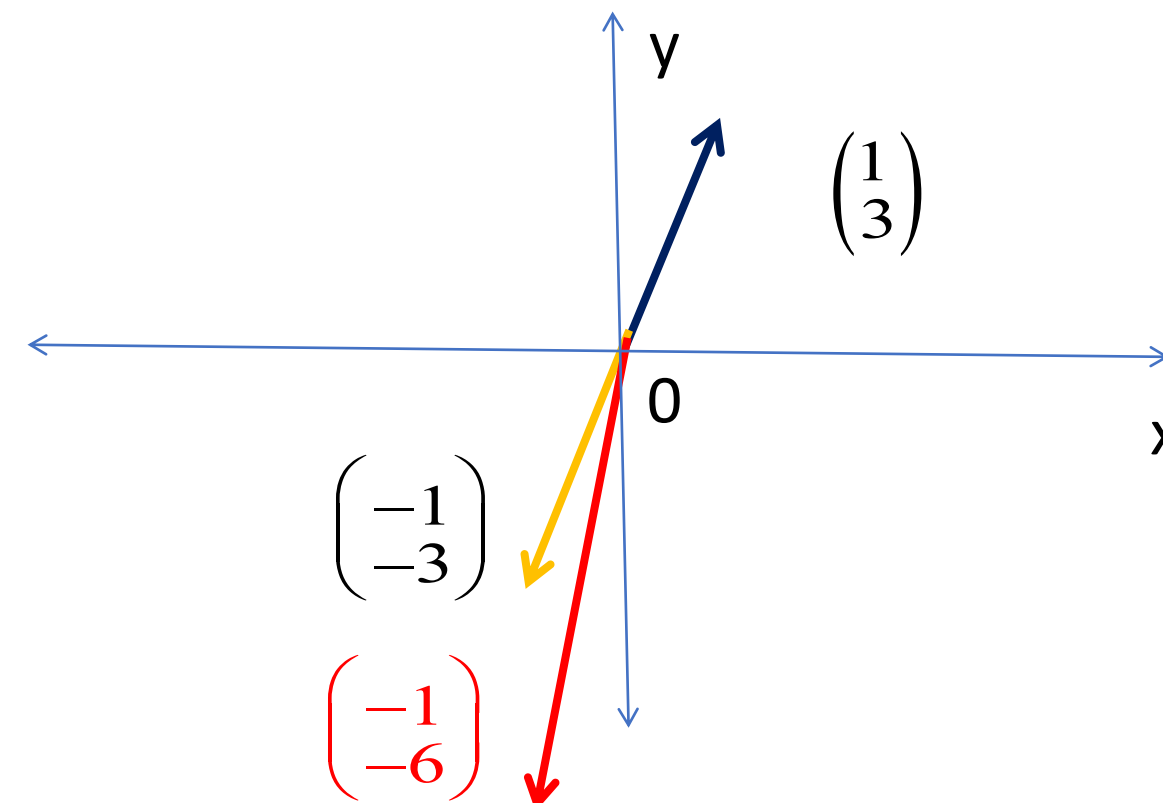
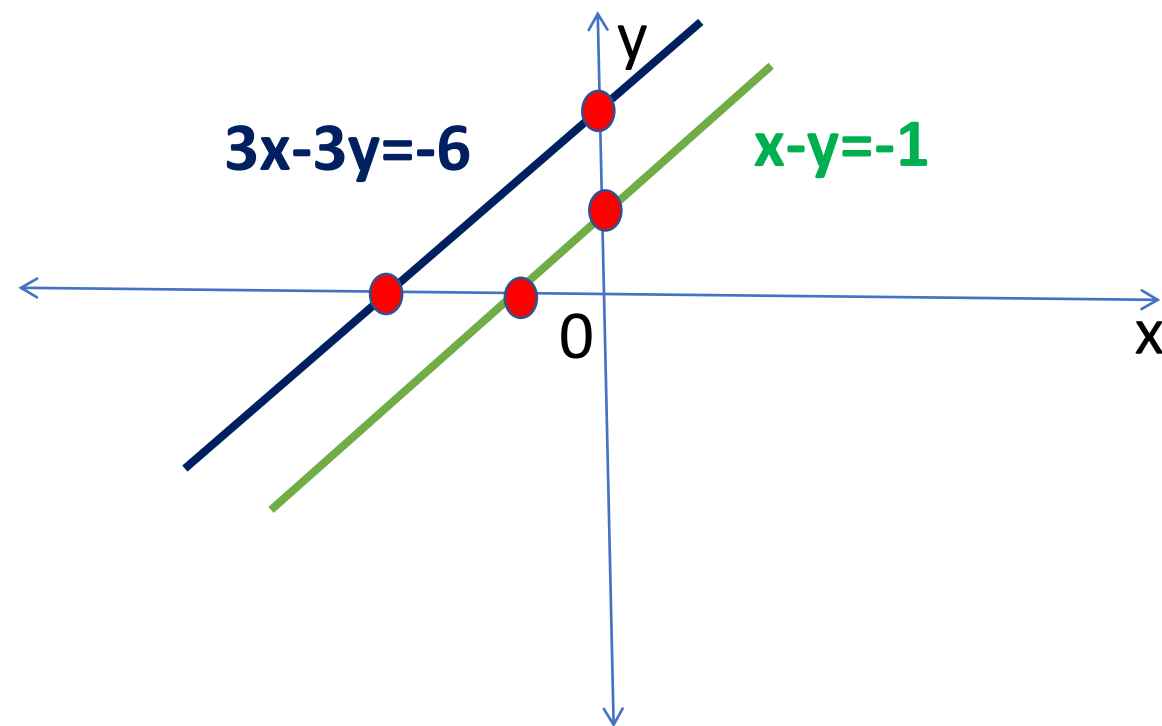
- (ii) Consider the system
- $$\begin{aligned}x - y &= -1 \\ 3x - 3y &= -6\end{aligned}$$

➤ **Row Picture:** These are parallel lines and so has no point of intersection. Hence this system has **no solution**.

❖ **Column Picture:**

$$x \begin{pmatrix} 1 \\ 3 \end{pmatrix} + y \begin{pmatrix} -1 \\ -3 \end{pmatrix} = \begin{pmatrix} -1 \\ -6 \end{pmatrix}$$

There is no linear combination of the column vectors on LHS which produces the vector on RHS.



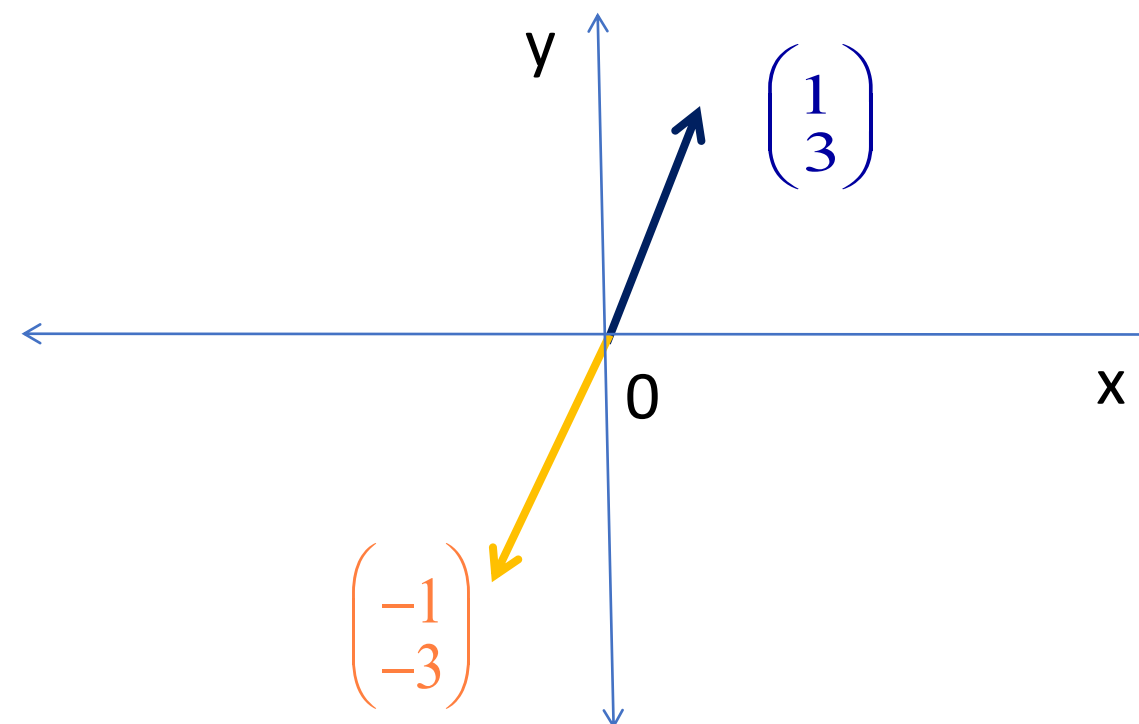
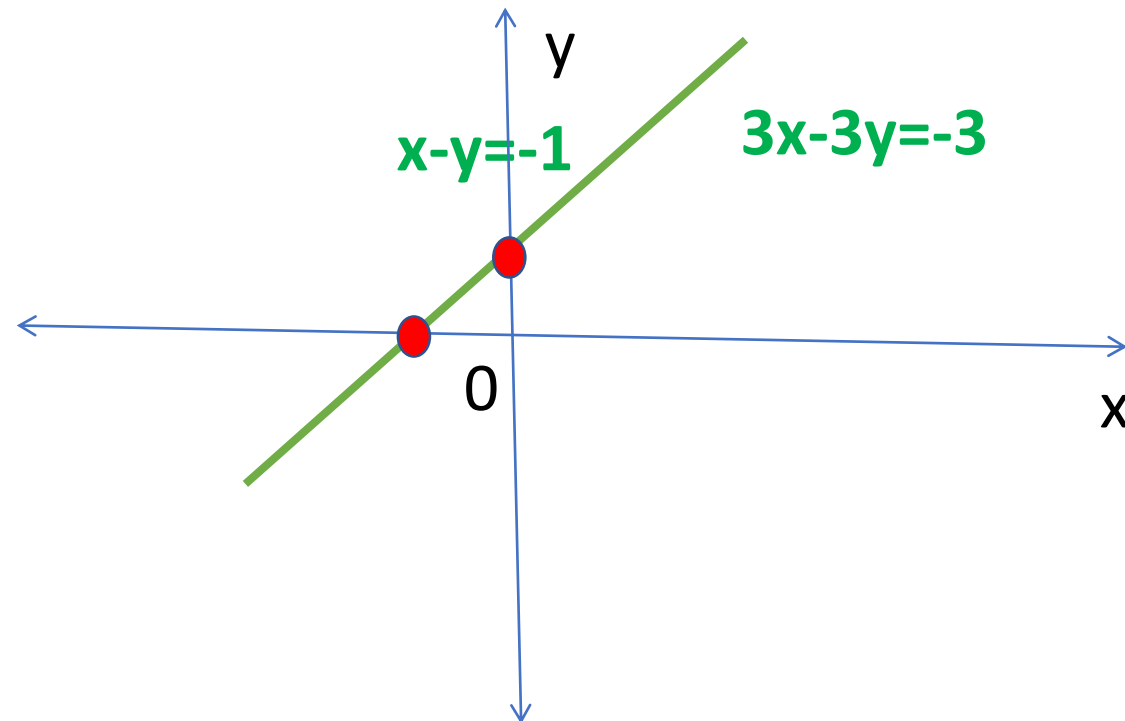
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THE GEOMETRY OF LINEAR EQUATIONS:

- (iii) Consider the system
- $$\begin{aligned}x - y &= -1 \\ 3x - 3y &= -3\end{aligned}$$
- **Row Picture:** These are coincident (overlapping) lines and so has infinite number of solutions. Hence this system has **infinitely many solutions**.

❖ **Column Picture:** $x \begin{pmatrix} 1 \\ 3 \end{pmatrix} + y \begin{pmatrix} -1 \\ -3 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$

There are infinite number of linear combinations of the column vectors on LHS which produces the vector on RHS.



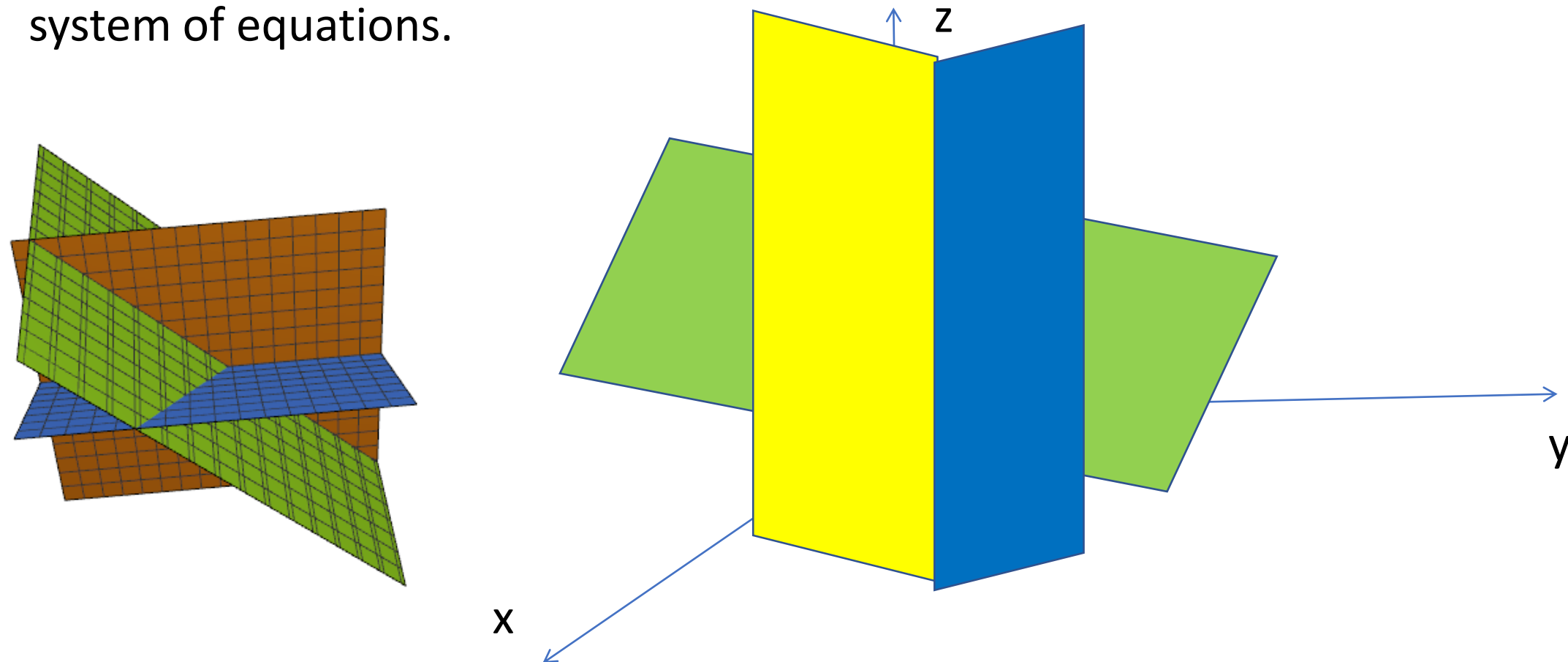
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THE GEOMETRY OF LINEAR EQUATIONS:

❖ **SYSTEM OF 3 EQUATIONS WITH 3 VARIABLES:** Consider the system

$$\begin{aligned}x + y + 2z &= 1 \\x + 2y - z &= -2 \\x + 3y + z &= 5\end{aligned}$$

❖ **Row Picture:** Each equation describes a 2-dimensional plane in $\mathbb{R}^3$. The first 2 planes intersect along a line and this line intersects with the third plane to produce a point $(-6, 3, 2)$ which is the unique solution (point of intersection of the three planes) to the system of equations.



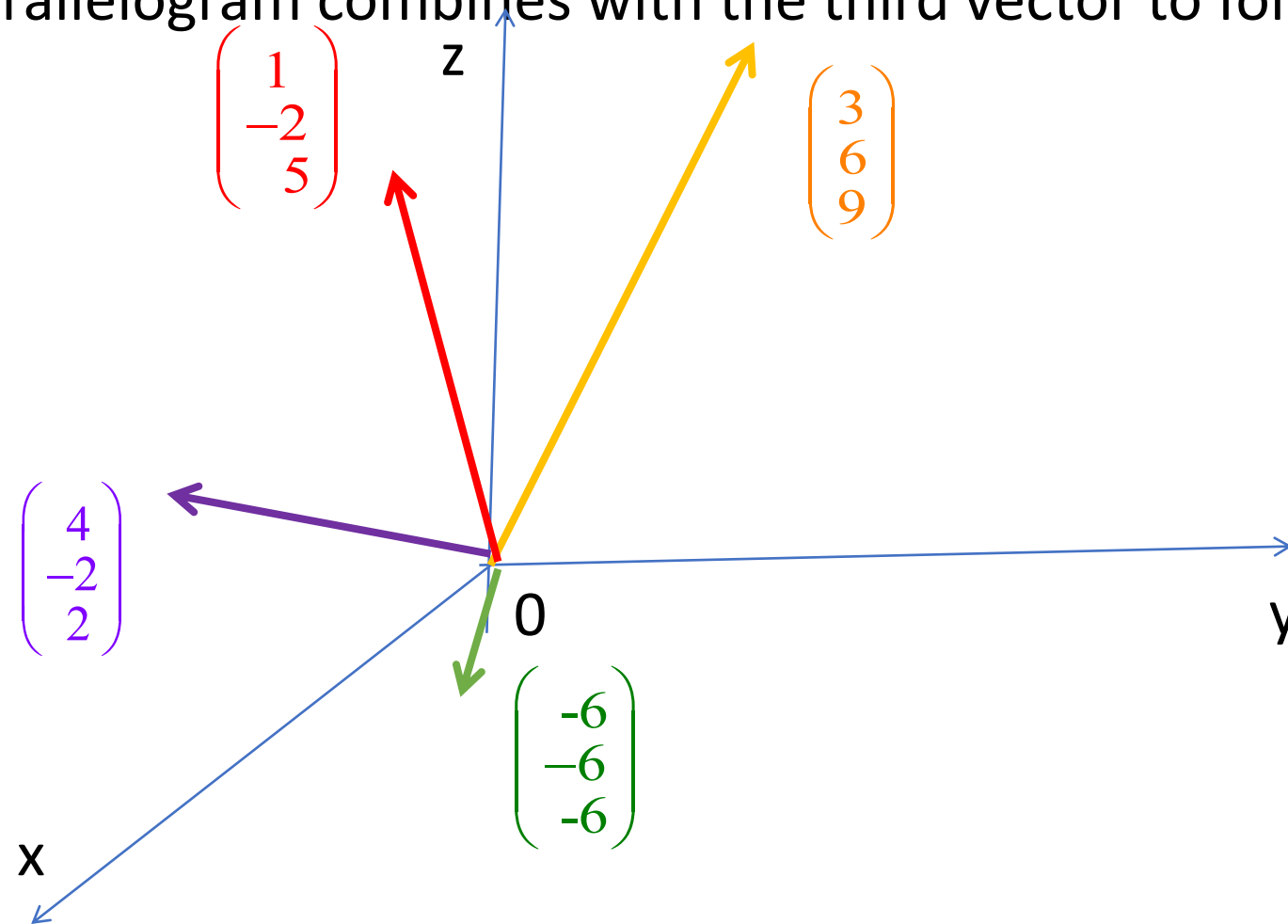
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THE GEOMETRY OF LINEAR EQUATIONS:

❖ **Column Picture:** The vector equation for the given system can be written as

$$x \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + z \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} \Rightarrow -6 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}$$

The first two vectors on the LHS combine to form a Parallelogram and the diagonal of this parallelogram combines with the third vector to form a Parallelepiped.



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THE GEOMETRY OF LINEAR EQUATIONS:

- ❖ **Row Picture:** Intersection of Lines/Planes.
- ❖ **Column Picture:** Combination of Columns.
- ❖ **Row Picture:** A Line requires 2 equations in 3-dimensional space. Similarly a Line requires 3 equations in 4-dimensional space. Hence a Line requires $(n-1)$ equations in n -dimensional space.
The first equation represents a $(n-1)$ dimensional plane in n dimensions. The second plane(equation) intersects it in a smaller set of dimension $(n-2)$. Thus every new plane reduces the dimension by one. At the end when all the ' n ' planes are accounted for the intersection has dimension zero. It is a point, which lies on all the planes and its co-ordinates satisfy all ' n ' equations. This point is the solution.



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References/Links:

https://upload.wikimedia.org/wikipedia/commons/c/c0/Intersecting_Lines.svg

Google search: Graphs of row and column picture for a system of linear equations





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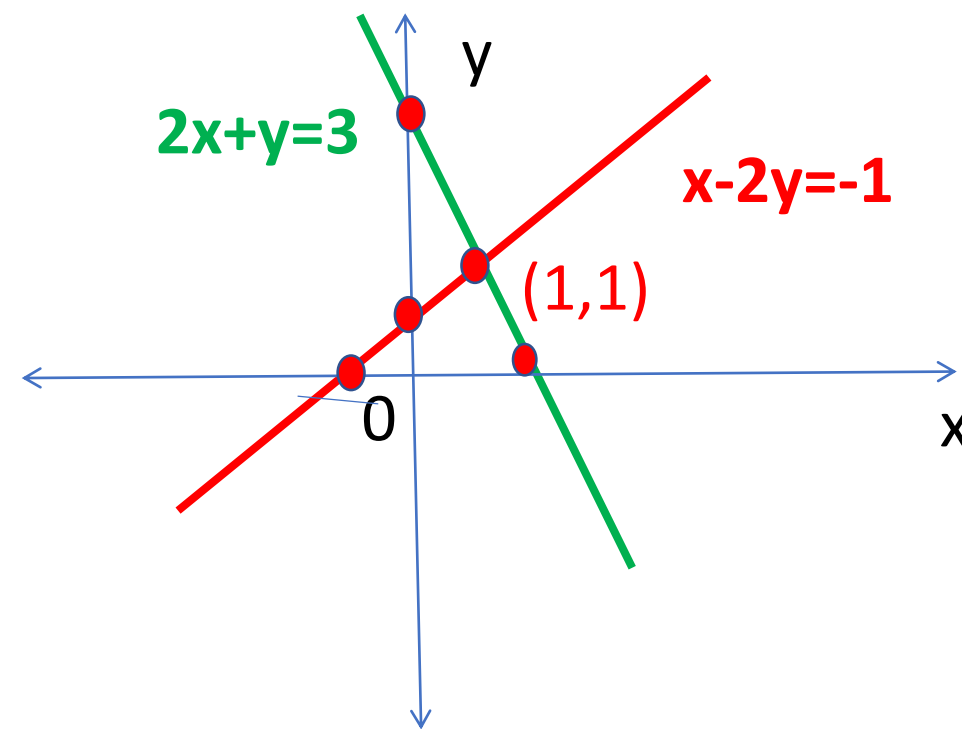
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THE GEOMETRY OF LINEAR EQUATIONS:

Course Content: The Geometry of Linear Equations

1. Solve the system of equations $2x+y=3$; $x-2y=-1$ and draw the Row picture and Column picture.

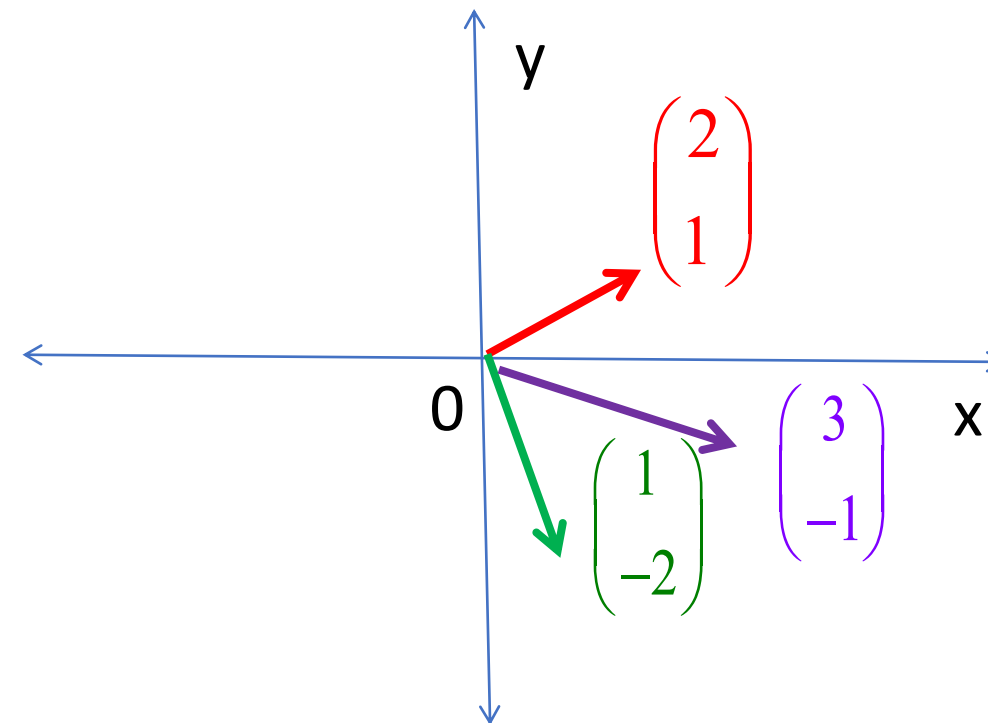
Row picture



Intersecting lines
Unique solution $(1,1)$

Column picture

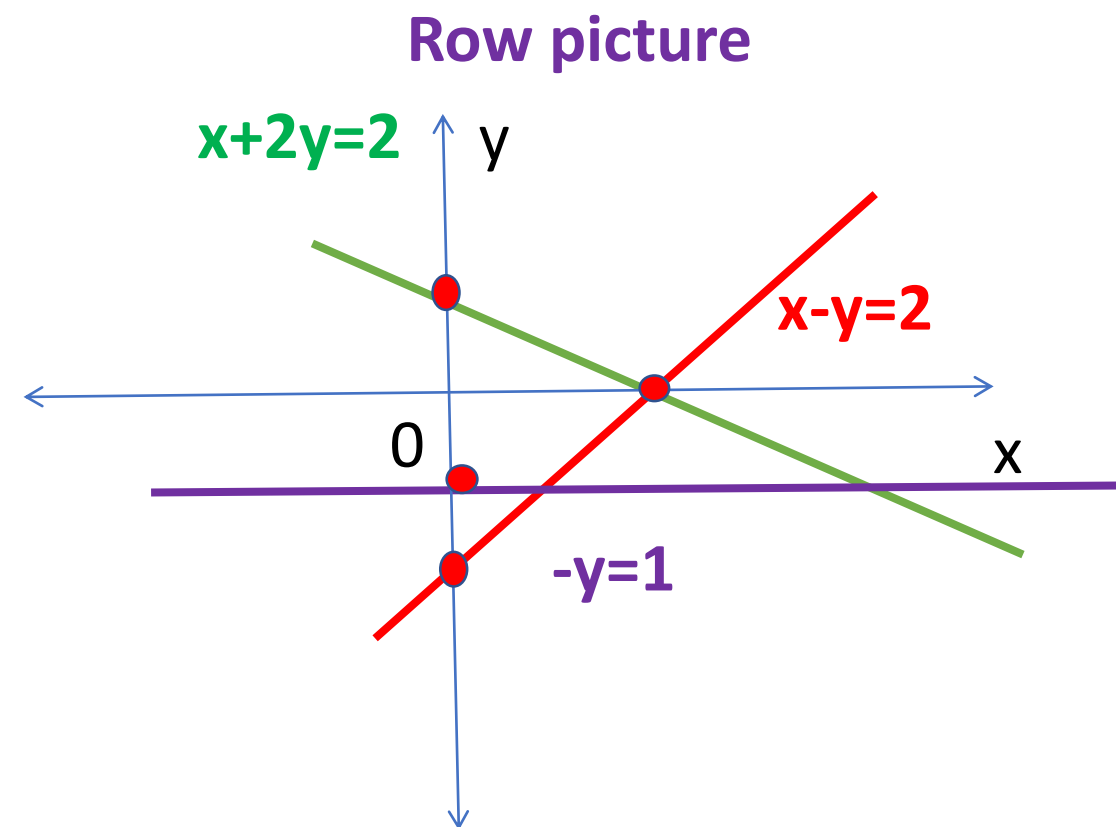
$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \Rightarrow 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$



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THE GEOMETRY OF LINEAR EQUATIONS:

2. Sketch the three lines $x+2y=2$; $x-y=2$; $-y=1$ (row picture only) and decide if the three equations are solvable. What happens if all right hand sides are zero? Is there any non-zero choice of right hand sides that allow the three lines to intersect at a common point of intersection?

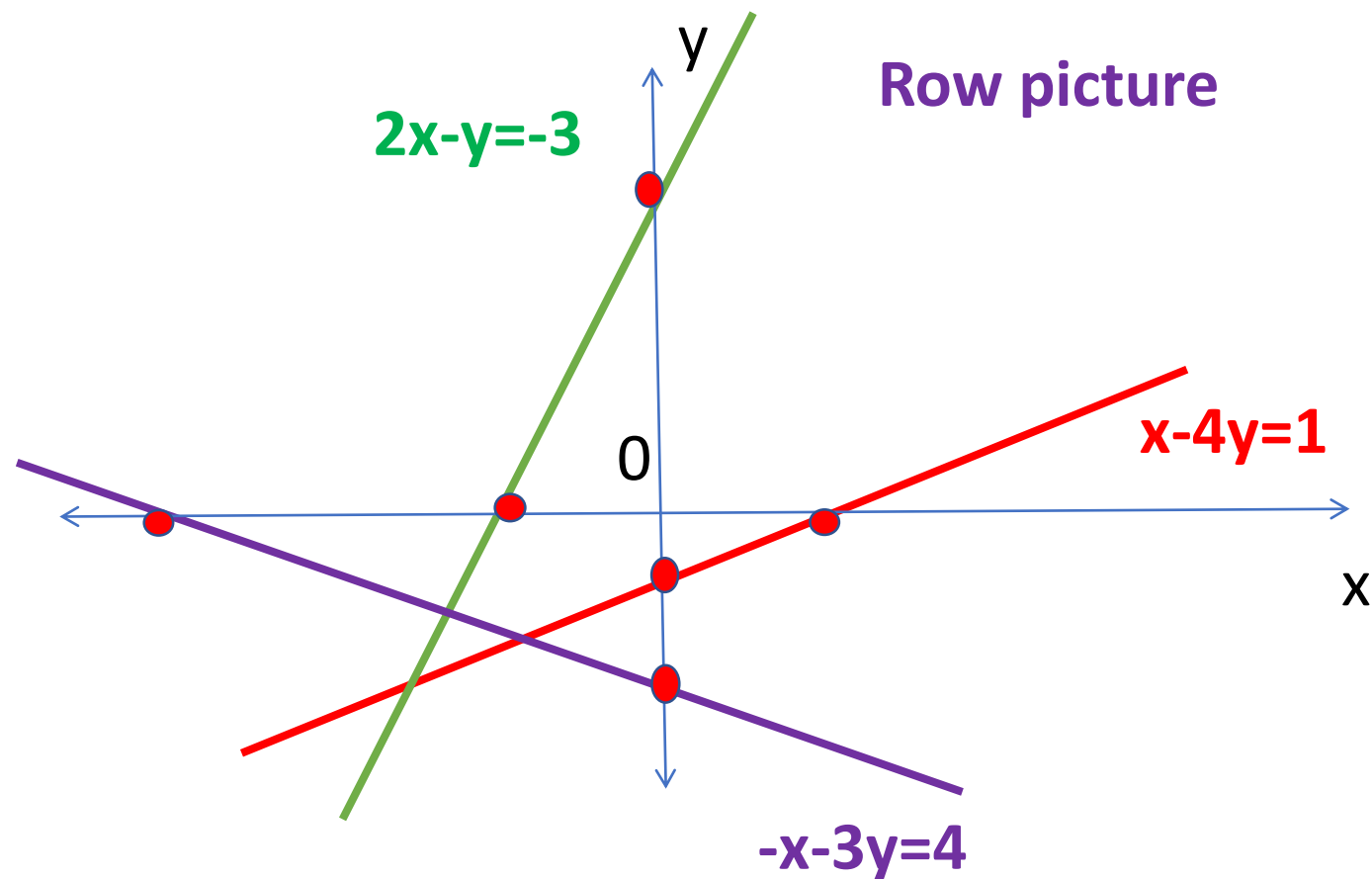


The given three equations are not solvable as they do not have a common point of intersection. If all right hand sides are zero we get a trivial(zero) solution i.e $x=0$, $y=0$. If RHS of first equation is -1 then the three lines $x+2y=-1$; $x-y=2$; $-y=1$ intersect at a common point of intersection $(1,-1)$

LINEAR ALGEBRA AND ITS APPLICATIONS

THE GEOMETRY OF LINEAR EQUATIONS:

3. Sketch the three lines $x - 4y = 1$; $2x - y = -3$; $-x - 3y = 4$ and decide if the three equations are solvable. Do they have a common point of intersection? Explain.



Given three equations are not solvable as they have no common point of intersection.

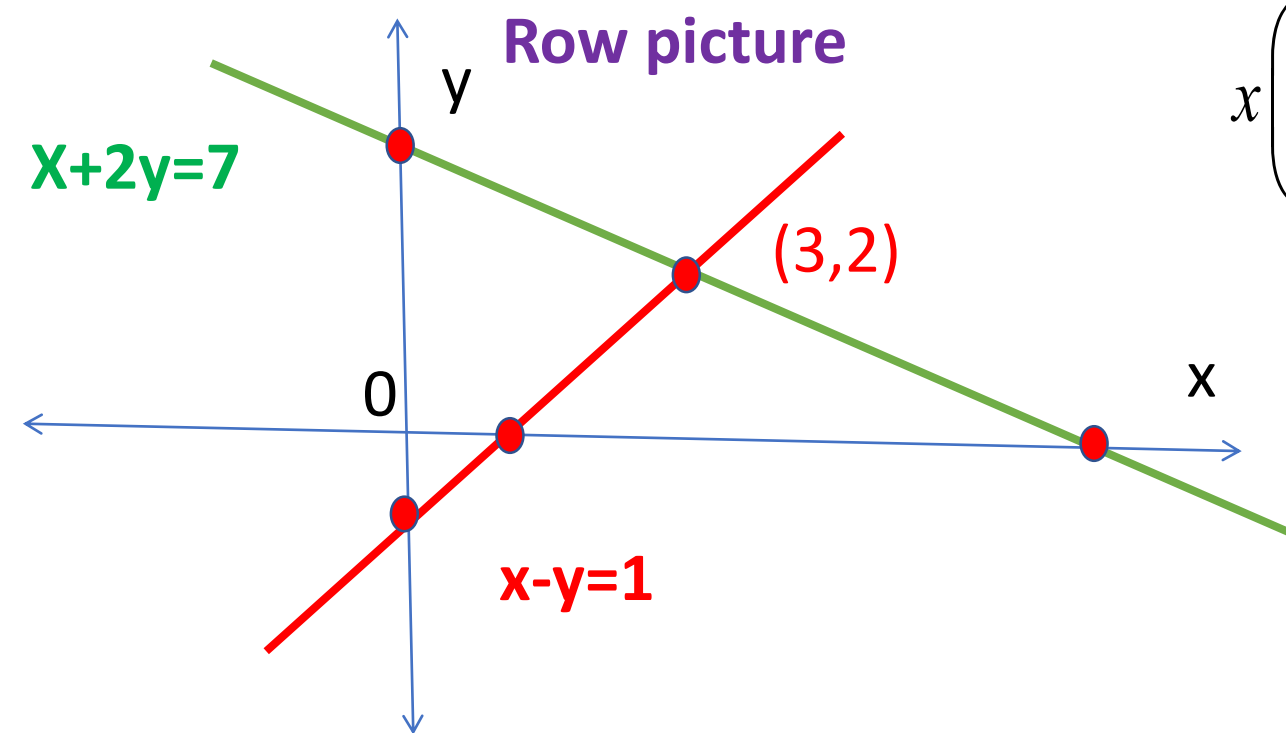
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THE GEOMETRY OF LINEAR EQUATIONS:

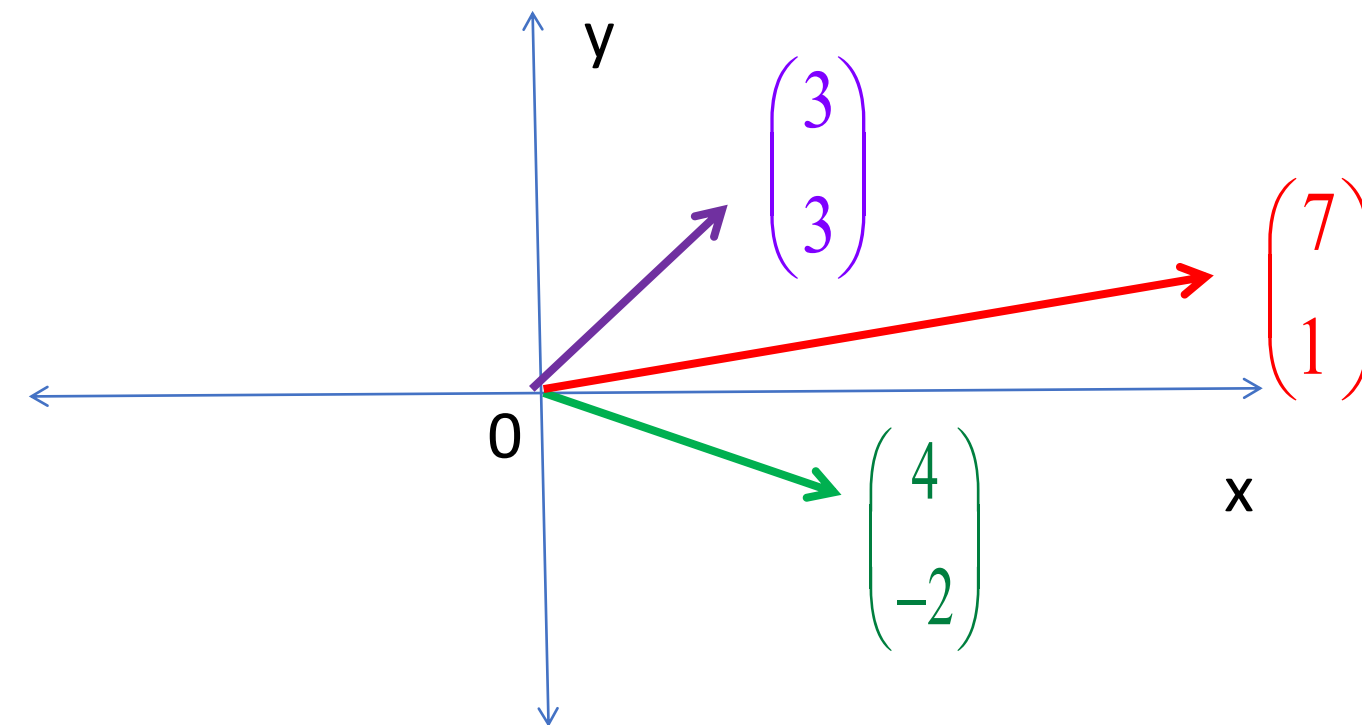
4. Draw the row picture and column picture for the following system of equations and discuss its consistency, singularity, and existence of the solution:

(i) $x + 2y = 7$; $x - y = 1$

Column picture



$$x \begin{pmatrix} 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix} \Rightarrow 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$$



(ii) $2x + 3y = 6$; $4x + 6y = 12$ (iii) $x - 3y = 5$; $x - 3y = -5$

LINEAR ALGEBRA AND ITS APPLICATIONS

INTRODUCTION:

Example:

$$A \rightarrow \left(\begin{array}{cccc|c} \textcircled{1} & 1 & -2 & 3 & 4 \\ 2 & 3 & 3 & -1 & 3 \\ 5 & 7 & 4 & -1 & 5 \end{array} \right) \xrightarrow[R_3 - 5R_1]{R_2 - 2R_1} \left(\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 2 & 14 & -16 & -15 \end{array} \right) \xrightarrow{R_3 - 2R_2}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 0 & 0 & -2 & -5 \end{array} \right) = U \xrightarrow{R_3 / (-2)} \left(\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 0 & 0 & 1 & 5/2 \end{array} \right) \xrightarrow[R_2 + 7R_3]{R_1 - 3R_3}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -2 & 0 & -7/2 \\ 0 & 1 & 7 & 0 & 25/2 \\ 0 & 0 & 0 & 1 & 5/2 \end{array} \right) \xrightarrow{R_1 - R_2} \left(\begin{array}{cccc|c} 1 & 0 & -9 & 0 & -16 \\ 0 & 1 & 7 & 0 & 25/2 \\ 0 & 0 & 0 & 1 & 5/2 \end{array} \right) = R$$

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References/Links:

https://upload.wikimedia.org/wikipedia/commons/c/c0/Intersecting_Lines.svg

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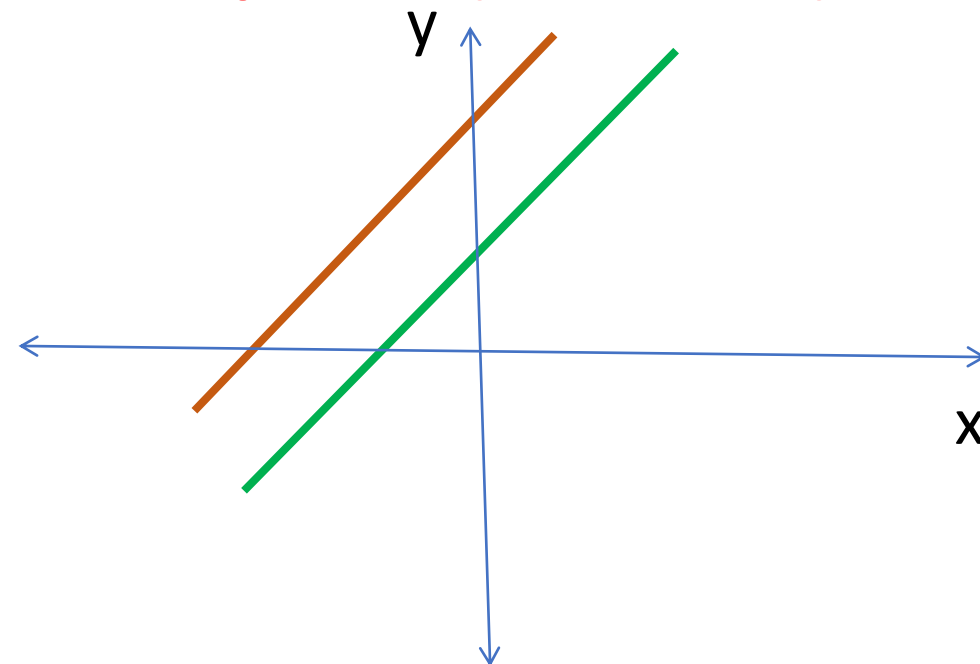
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Course Content: Singular Cases

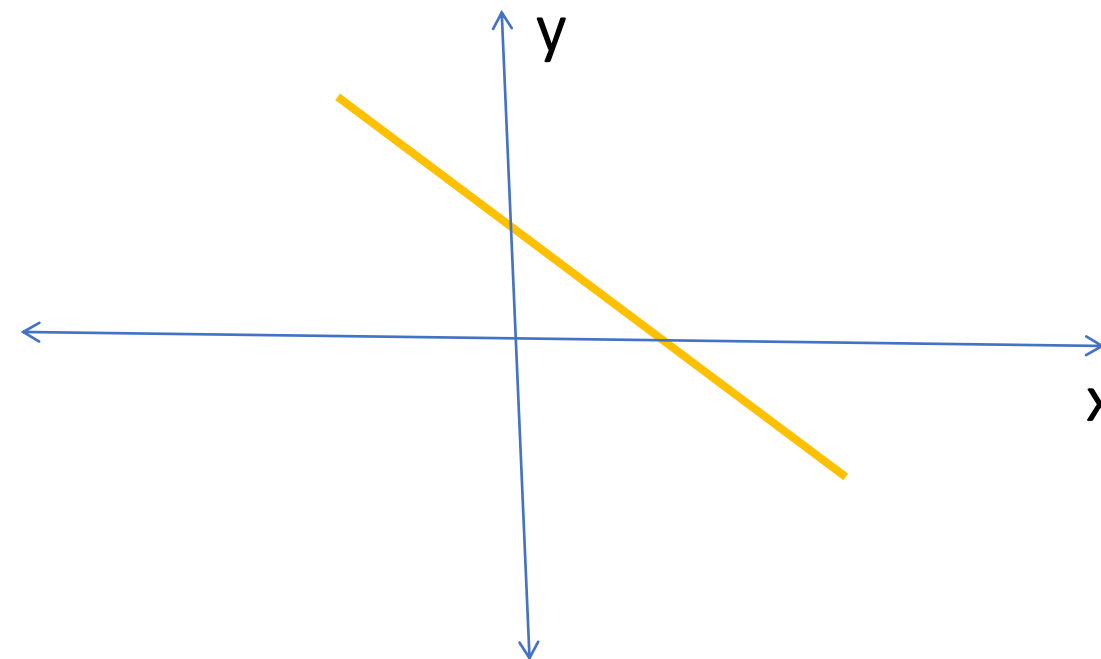
- ❖ **SINGULAR CASES in Two dimensions:** A system of linear equations is said to be singular ($|A| = 0$) if it has no solution or has infinite number of solutions.
- **(i) ROW PICTURE (Two dimensions):** In two dimensions the lines are parallel if they have no solution and coincident if they have infinite number of solutions. In such a case the matrix A will have dependent row /column and $\det(A)=0$. Such a matrix is called Singular Matrix.

(2 lines):

Lines parallel (no solution)



Lines coincident (infinite no. of solutions)

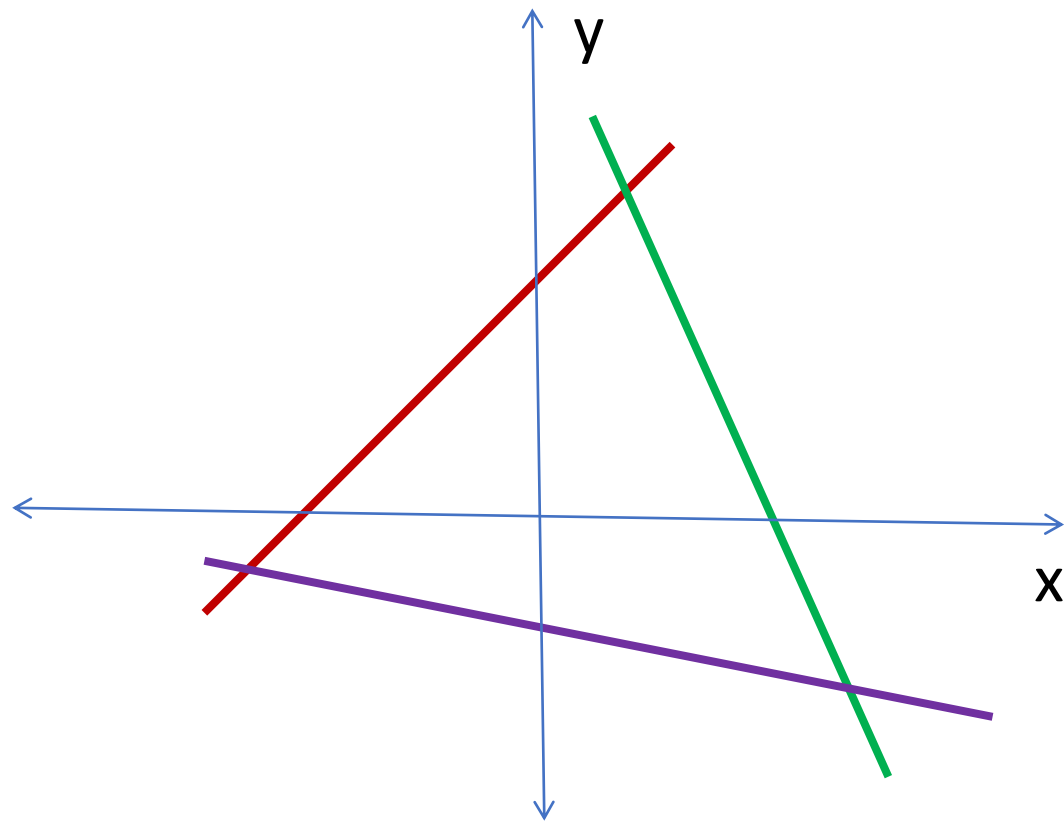


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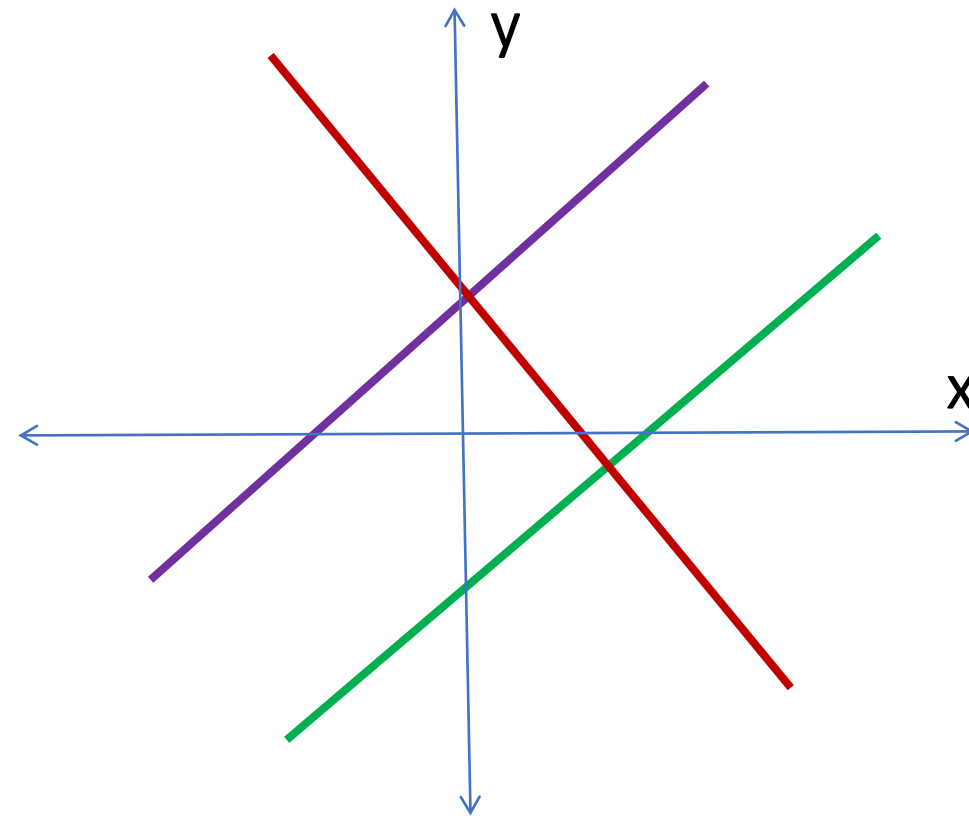
THE GEOMETRY OF LINEAR EQUATIONS:

(3 lines):

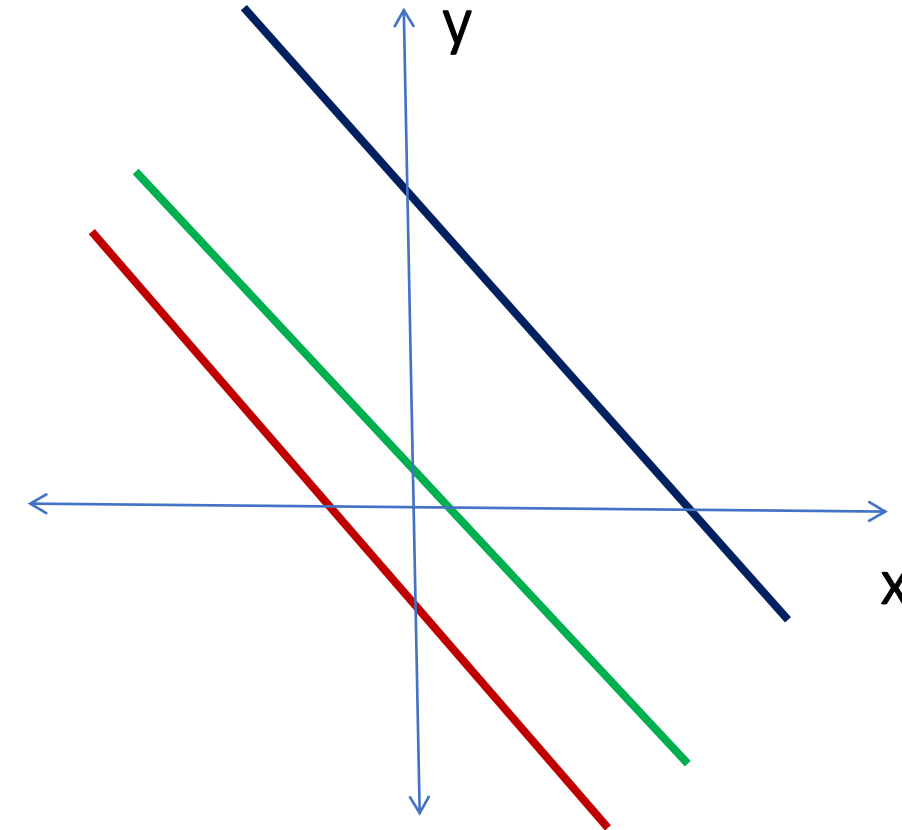
Lines intersecting in pairs
(no solution)



2 Lines parallel & one
intersecting (no solution)



All 3 parallel lines
(no solution)



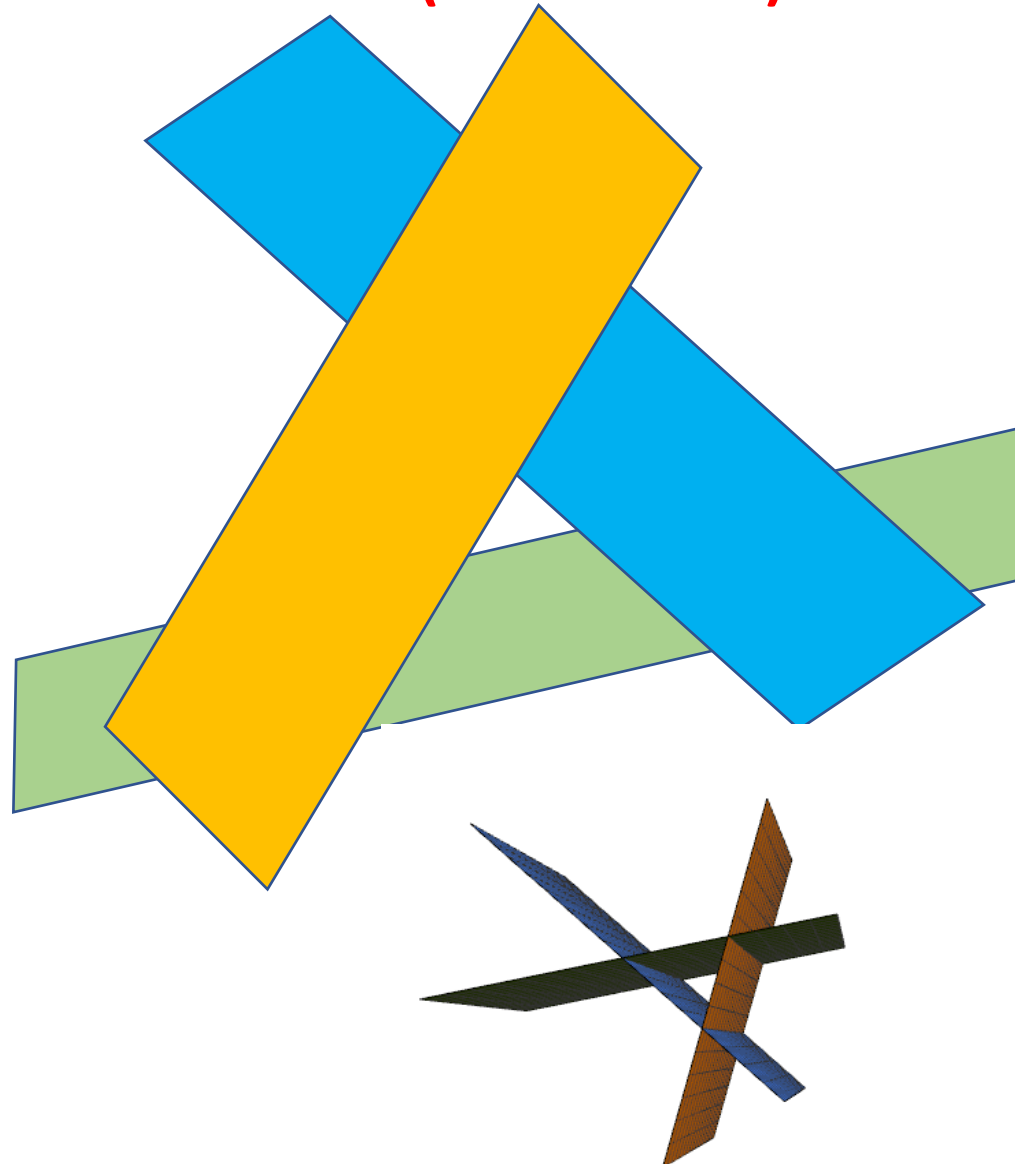
LINEAR ALGEBRA AND ITS APPLICATIONS

THE GEOMETRY OF LINEAR EQUATIONS:

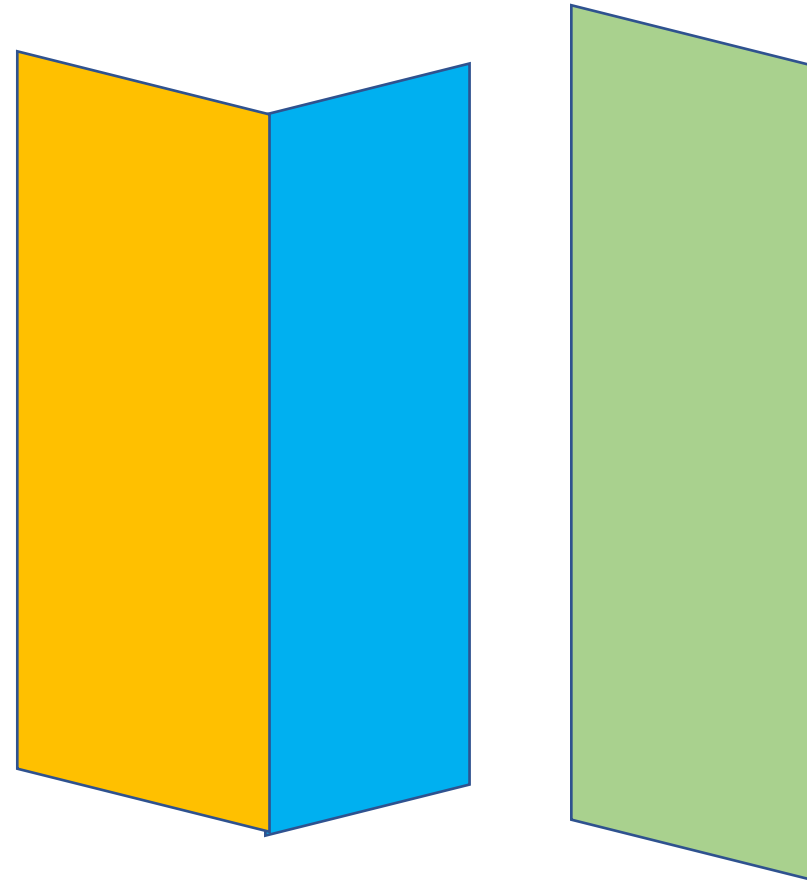
- **ROW PICTURE(Three dimensions):** In three dimensions if the 3 Planes do not intersect then we have the following cases :

(3 planes):

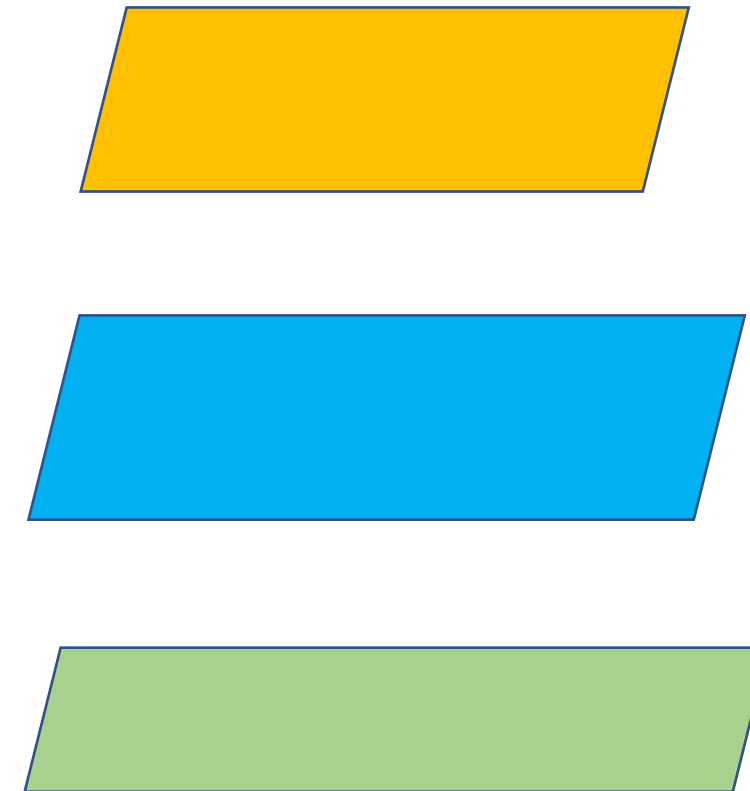
Every pair of planes intersecting in a line (no solution)



2 planes intersecting in a line and 3rd is parallel to this line (no solution)



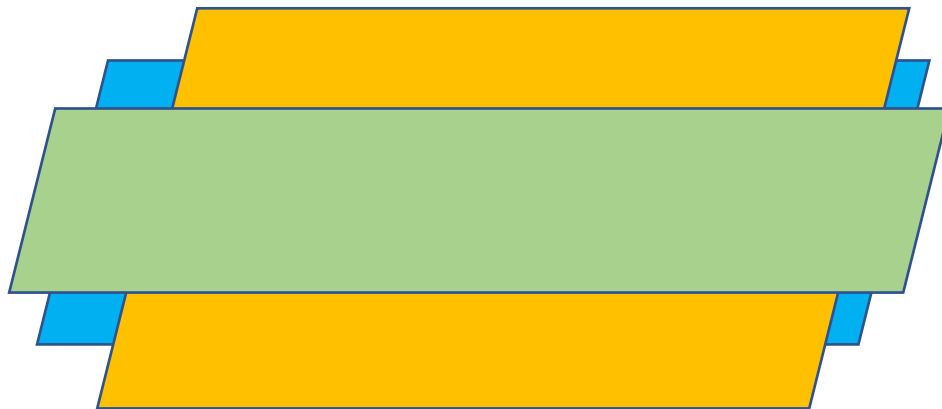
All 3 parallel planes (no solution)



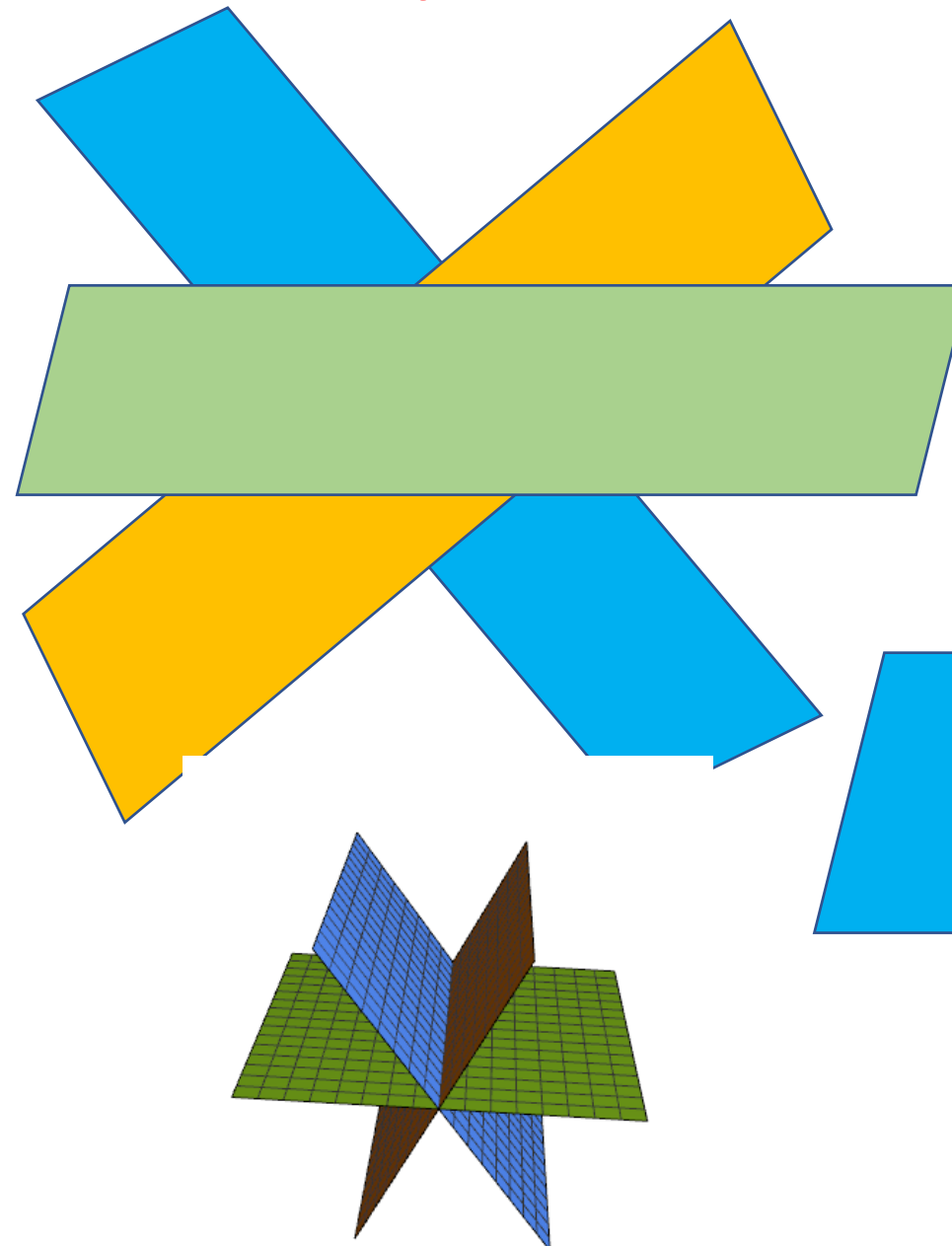
LINEAR ALGEBRA AND ITS APPLICATIONS

THE GEOMETRY OF LINEAR EQUATIONS:

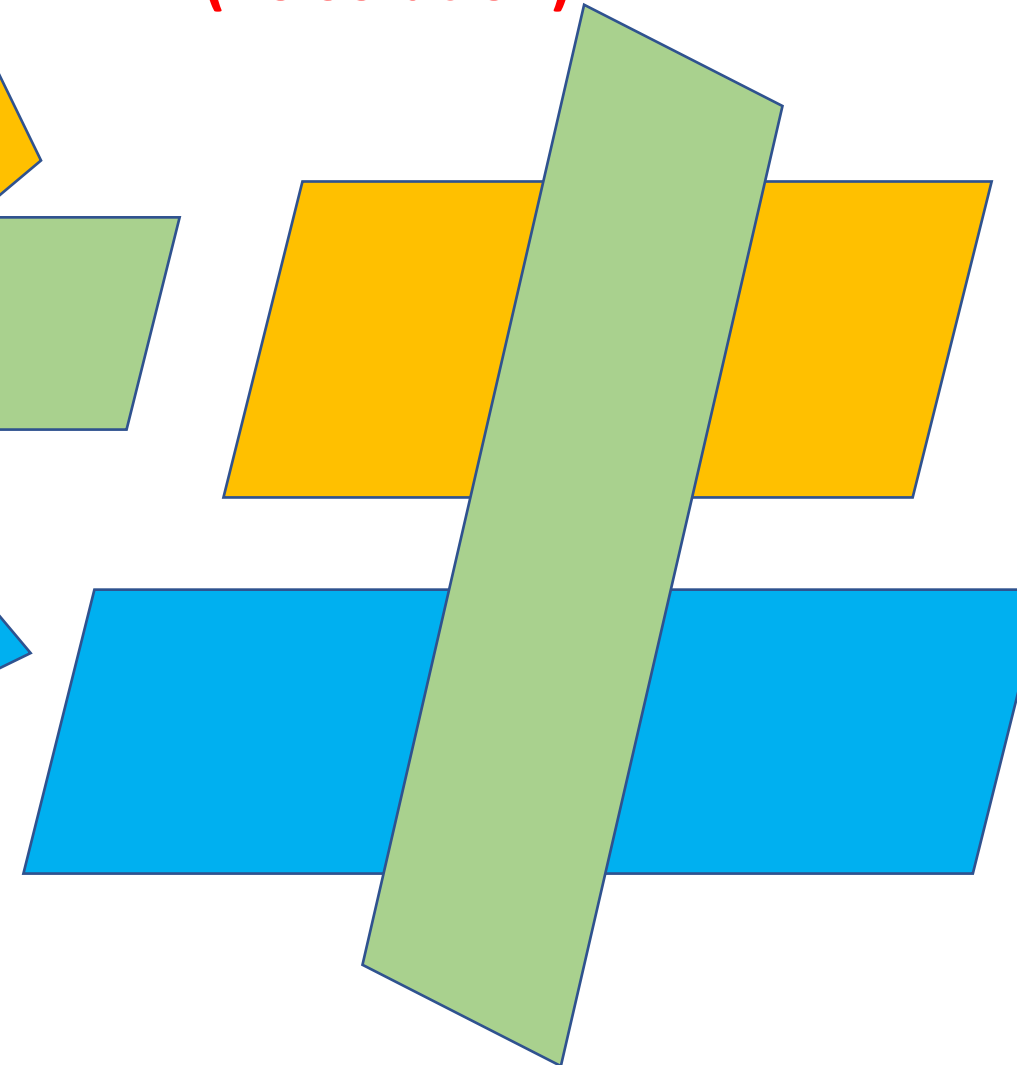
All 3 overlapping planes
(infinite no. of solutions)



All 3 planes intersecting
in a line (infinite no
solutions)



2 parallel planes
and third plane
intersecting them
(no solution)



LINEAR ALGEBRA AND ITS APPLICATIONS

THE GEOMETRY OF LINEAR EQUATIONS:

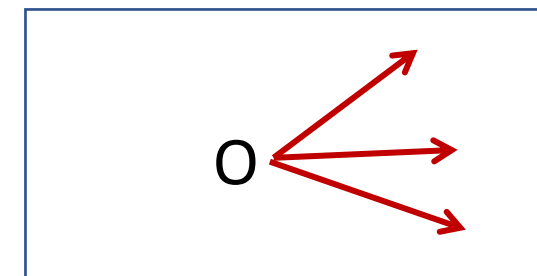
- (ii) **COLUMN PICTURE (Three dimensions)**: Consider the column picture for a system of 3 equations in 3 variables

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \Rightarrow x \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \end{pmatrix} + y \begin{pmatrix} a_{12} \\ a_{22} \\ a_{32} \end{pmatrix} + z \begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

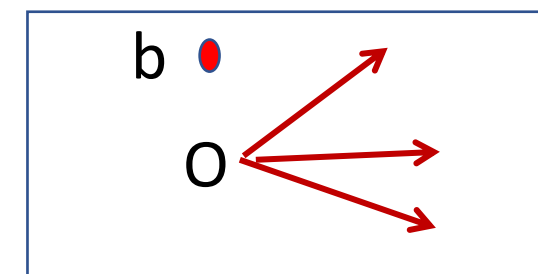
Each of these column vectors is a position vector with origin as a point, These column vectors lie in a plane (as they pass through the origin). Then every combination of these vectors on LHS lie in the same plane (3 vectors are coplanar).

• b

If the vector b is not in that plane, then solution is impossible. This system is singular and has no solution.



If the vector b lies in the plane, (i.e b is also coplanar), then there are too many solutions. The 3 columns combine in infinitely many ways to produce b. This system is singular and has infinite no. of solutions.



ENGINEERING MATHEMATICS-III

References/Links:

https://upload.wikimedia.org/wikipedia/commons/c/c0/Intersecting_Lines.svg

Google search: Graphs of row and column picture for a system of linear equations





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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Gaussian Elimination

❖ **Rank Of A Matrix:** A Square matrix A of order n is said to have rank r if

- At least one minor of order r does not vanish ($\neq 0$).
- Every minor of order (r+1) vanishes ($= 0$).

Rank of matrix A is denoted by r i.e. rank(A)=r.

- ❖ If $A = [a_{ij}]_{m \times n}$ is a rectangular matrix, then **Rank of the Matrix** is defined as the number of non-zero rows in the Echelon form of A. It is also defined as the maximum number of Linearly Independent Rows or Columns of the Matrix A.

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 0 \end{pmatrix}; \begin{pmatrix} 2 & 2 & -3 \\ -2 & -2 & 3 \\ 4 & 4 & 6 \end{pmatrix}; \begin{pmatrix} 1 & 5 \\ -1 & 3 \\ 2 & 3 \end{pmatrix}; \begin{pmatrix} 1 & 5 \\ -1 & 3 \\ 2 & 10 \end{pmatrix}; \begin{pmatrix} 2 & -1 & 3 & 5 \\ 3 & 2 & 1 & 2 \end{pmatrix}$$

$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$

2 1 2 2 2

Ex: Find the conditions on a and b so that the Matrix $\begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 1 & 3 & 6 \end{pmatrix}$ has rank 1, 2, 3.

$$\begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 1 & 3 & 6 \end{pmatrix} \xrightarrow{R_3 - \left(\frac{1}{a}\right)R_1} \begin{pmatrix} a & 1 & 2 \\ 0 & 2 & b \\ 0 & 3 - (1/a) & 6 - (2/a) \end{pmatrix}$$

- (i) For no values of a and b this matrix will have rank 1.
- (ii) If $a=1/3$ and $b=4$, rank of the matrix is 2.
- (iii) If $a \neq 1/3$ and $b \neq 4$, rank of the matrix is 3.

GAUSSIAN ELIMINATION:

❖ Relation between Rank, Consistency and Solution:

❖ If $\text{rank}(A)=r$, then the following hold good:

- (i) If $\text{rank}(A)=\text{rank}[A:b]=r$, system $Ax=b$ is **consistent** and has **a solution**.
- (ii) If $\text{rank}(A)=\text{rank}[A:b]=r=n$, system $Ax=b$ is **consistent** and has **a unique solution**.
- (iii) If $\text{rank}(A)=\text{rank}[A:b]=r<n$, system $Ax=b$ is **consistent** and has **infinite number of solutions**.
- (iv) If $\text{rank}(A) \neq \text{rank}[A:b]$, system $Ax=b$ is **inconsistent** and has **no solution**.

Gaussian Elimination

Gaussian Elimination is used to check for Consistency and solve a System of linear equations.

For a given system of equations $Ax=b$ apply Elementary row transformations to the Augmented Matrix $[A:b]$ and reduce it to $[U:c]$ where U is an **Upper Triangular Matrix** so that we get an equivalent system $Ux=c$ which can be solved by **Backward Substitution**.

Here A and U are Equivalent Matrices and hence solution of $Ax=b$ is same as $Ux=c$.

❖ **The following steps are to be followed while performing Elementary Row Transformations in Gaussian Elimination:**

- **No exchange** of rows.
- First row should be **retained** as it is (**not altered**).
- The first non-zero element in every non-zero row is called **Pivot**.
- The original system $Ax=b$ and new system obtained $Ux=c$ have the **same solution**.

GAUSSIAN ELIMINATION:

Gaussian Elimination is illustrated below for a system of 3 equations with 3 variables.

Consider a system of 3 equations in 3 variables $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$\begin{aligned} [A : b] &= \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ a_{21} & a_{22} & a_{23} : b_2 \\ a_{31} & a_{32} & a_{33} : b_3 \end{pmatrix} \xrightarrow[R_3 - \left(\frac{a_{31}}{a_{11}}\right)R_1]{R_2 - \left(\frac{a_{21}}{a_{11}}\right)R_1} \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ 0 & d_{22} & d_{23} : c_2 \\ 0 & d_{32} & d_{33} : c_3 \end{pmatrix} \\ &\xrightarrow{R_3 - \left(\frac{d_{32}}{d_{22}}\right)R_2} \begin{pmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ 0 & d_{22} & d_{23} : c_2 \\ 0 & 0 & e_{33} : c_4 \end{pmatrix} = [U : c] \\ &\Rightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ d_{22}x_2 + d_{23}x_3 = c_2 \\ e_{33}x_3 = c_4 \end{cases} \end{aligned}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:

Check for consistency and solve the following system of equations if consistent:

$$\begin{aligned} \text{(i)} \quad & x_1 + x_2 - 2x_3 + 4x_4 = 5 \\ & 2x_1 + 2x_2 - 3x_3 + x_4 = 3 \\ & 3x_1 + 3x_2 - 4x_3 - 2x_4 = 1 \end{aligned} \quad [A:b] = \begin{pmatrix} 1 & 1 & -2 & 4 : 5 \\ 2 & 2 & -3 & 1 : 3 \\ 3 & 3 & -4 & -2 : 1 \end{pmatrix}$$

$$\xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 1 & -2 & 4 : 5 \\ 0 & 0 & 1 & -7 : -7 \\ 0 & 0 & 2 & -14 : -14 \end{bmatrix} \xrightarrow{R_3 - 2R_2} \begin{bmatrix} 1 & 1 & -2 & 4 : 5 \\ 0 & 0 & 1 & -7 : -7 \\ 0 & 0 & 0 & 0 : 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + x_2 - 2x_3 + 4x_4 = 5 \\ x_3 - 7x_4 = -7 \end{cases} \quad \begin{matrix} r(A)=2 = r[A:b] < n(=4) \\ \text{System is consistent and has infinitely many solutions.} \end{matrix}$$

Solution is $(x_1, x_2, x_3, x_4) = (10k_1 - k_2 - 9, k_2, 7k_1 - 7, k_1)$

Depending upon values of k_1 and k_2 we get infinity of solutions.



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MATRICES AND GAUSSIAN ELIMINATION

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LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:

1. Check for consistency and solve the following system of equations if consistent:

$$(i) \quad \begin{aligned} x_1 + x_2 - 2x_3 + 3x_4 &= 4 \\ 2x_1 + 3x_2 + 3x_3 - x_4 &= 3 \\ 5x_1 + 7x_2 + 4x_3 + x_4 &= 5 \end{aligned} \quad [A:b] = \left(\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 2 & 3 & 3 & -1 & 3 \\ 5 & 7 & 4 & 1 & 5 \end{array} \right) \xrightarrow{\substack{R_2 - 2R_1 \\ R_3 - 5R_1}}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 2 & 14 & -14 & -15 \end{array} \right] \xrightarrow{\substack{R_3 - 2R_2}} \left[\begin{array}{cccc|c} 1 & 1 & -2 & 3 & 4 \\ 0 & 1 & 7 & -7 & -5 \\ 0 & 0 & 0 & 0 & -5 \end{array} \right]$$

This gives $0 = -5$ which is not possible.

Also $r(A) = 2$ and $r[A:b] = 3$

System is **inconsistent** and has **no solution**

GAUSSIAN ELIMINATION:

$$\begin{aligned} \text{(ii)} \quad & x_1 + 2x_2 + x_3 = 3 \\ & 2x_1 + 5x_2 - x_3 = -4 \\ & 3x_1 - 2x_2 - x_3 = 5 \end{aligned}$$

$$[A:b] = \begin{pmatrix} 1 & 2 & 1 & : & 3 \\ 2 & 5 & -1 & : & -4 \\ 3 & -2 & -1 & : & 5 \end{pmatrix}$$

$$\downarrow \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 1 & -3 & : & -10 \\ 0 & -8 & -4 & : & -4 \end{bmatrix}$$

$$\downarrow \begin{matrix} R_3 + 8R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 1 & -3 & : & -10 \\ 0 & 0 & -28 & : & -84 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 3 \\ x_2 - 3x_3 = -10 \\ -28x_3 = -84 \end{cases}$$

$r(A)=r[A:b]=3=n$. System is **consistent** and has **a unique solution**.

$$(x_1, x_2, x_3) = (2, -1, 3)$$

GAUSSIAN ELIMINATION:

$$\begin{aligned} \text{(iii)} \quad & 2x - 3y + 2z = 1 \\ & 5x - 8y + 7z = 1 \\ & y - 4z = 3 \end{aligned}$$

$$\begin{pmatrix} 2 & -3 & 2:1 \\ 5 & -8 & 7:1 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{\substack{R_2 \leftarrow \begin{pmatrix} 5 \\ -8 \\ 7 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix} R_1 \\ R_3 \leftarrow \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} R_2}} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & -1/2 & 2:-3/2 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & 1 & -4:3 \\ 0 & -1/2 & 2:-3/2 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2x - 3y + 2z = 1 \\ -(1/2)y + 2z = -3/2 \end{cases}$$

$r(A)=r(A:b)=2 < n(=3)$ hence system is **consistent**
and has **infinite number of solutions**.
i.e $(x, y, z) = (5k+5, 4k-+3, k)$

GAUSSIAN ELIMINATION:

2. Find all values of a for which the resulting linear system has (a) no solution (b) a unique solution and (c) infinitely many solutions:

$$x + y - z = 2$$

$$x + 2y + z = 3$$

$$x + y + (a^2 - 5)z = a$$

$$[A:b] = \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 1 & 2 & 1 & : & -3 \\ 1 & 1 & a^2 - 5 & : & a \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 0 & 1 & 2 & : & -5 \\ 0 & 0 & a^2 - 4 & : & a - 2 \end{pmatrix}$$

- (a) System has **no solution** if $a = -2$ (when $r(A) \neq r(A:b)$)
- (b) System has **a unique solution** if $a \neq \pm 2$ (when $r(A) = r(A:b) = 3 = n$)
- (c) System has **infinitely many solutions** if $a = 2$ (when $r(A) = r(A:b) = 2 < n$)

GAUSSIAN ELIMINATION:

3. Find an equation relating a , b and c so that the linear system
- $$\begin{aligned} x + 2y - 3z &= a \\ 2x + 3y + 3z &= b \\ 5x + 9y - 6z &= c \end{aligned}$$
- is consistent for any values of a , b and c that satisfy that equation. When $(a,b,c)=(2,3,9)$, then what is the solution of the system.

$$\begin{pmatrix} 1 & 2 & -3 : a \\ 2 & 3 & 3 : b \\ 5 & 9 & -6 : c \end{pmatrix} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 5R_1}]{\downarrow} \begin{pmatrix} 1 & 2 & -3 : a \\ 0 & -1 & 9 : b - 2a \\ 0 & -1 & 9 : c - 5a \end{pmatrix}$$

$$\xrightarrow[\substack{R_3 - R_2}]{\downarrow} \begin{pmatrix} 1 & 2 & -3 : a \\ 0 & -1 & 9 : b - 2a \\ 0 & 0 & 0 : c - b - 3a \end{pmatrix}$$

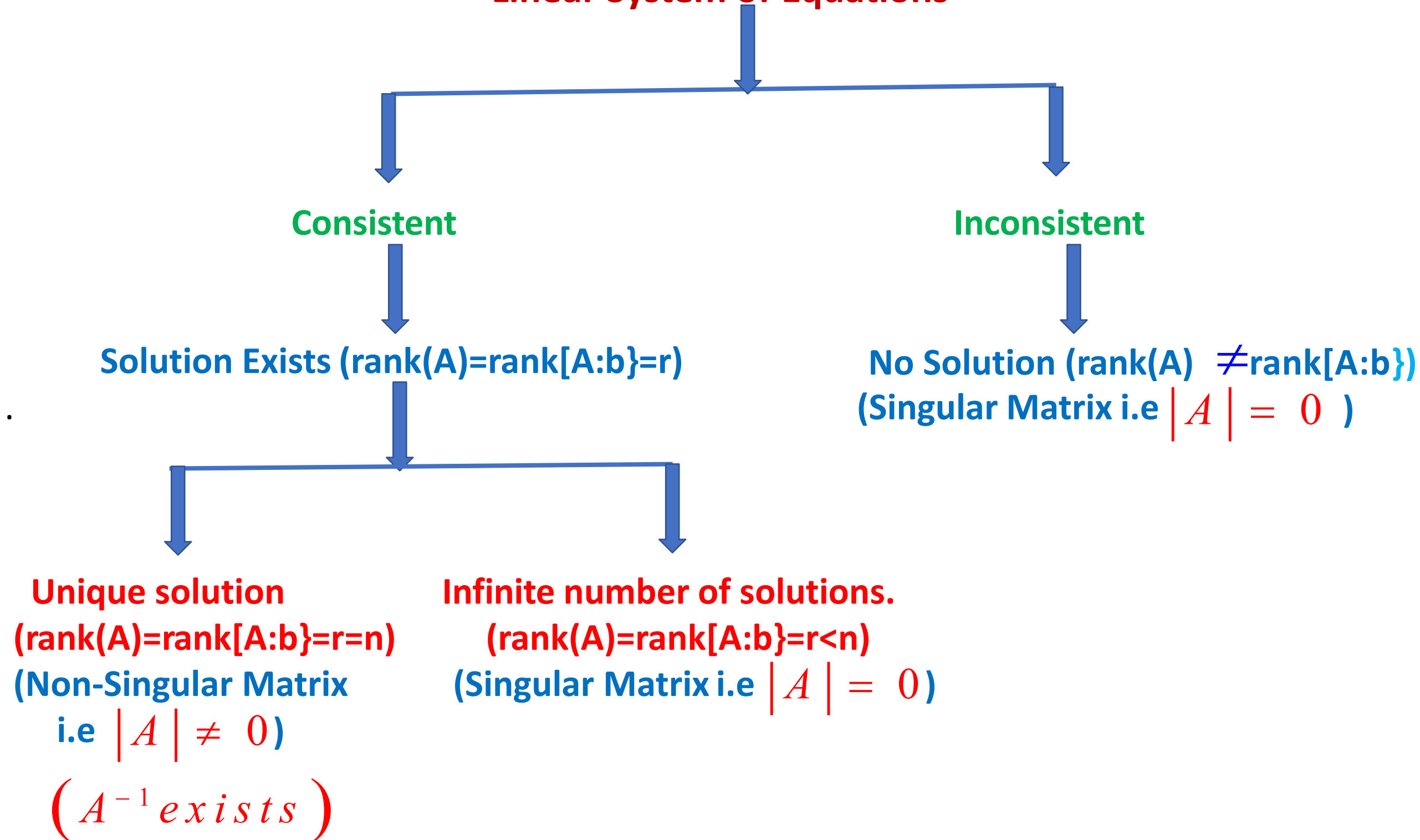
The given linear system will be **consistent** if a, b, c satisfy the relation $c - b - 3a = 0$.

When $(a, b, c) = (2, 3, 9)$, then the solution of the system is $(x, y, z) = (-15k, 9k + 1, k)$

LINEAR ALGEBRA AND ITS APPLICATIONS

GAUSSIAN ELIMINATION:

Linear System of Equations





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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Breakdown of Gaussian Elimination

- ❖ If a zero appears in a pivot position, elimination has to stop –either temporarily or permanently. In this case the system may or may not be singular.
In many cases this problem can be cured and elimination can proceed. Such a system is non-singular and has a full set of pivots.
In other cases, when the breakdown is unavoidable (permanent). These systems are singular and have no solution or have infinitely many solutions. For such systems a full set of pivots cannot be found.
- **Non-Singular & Curable** ($|A| \neq 0$)
- **Singular & InCurable** ($|A| = 0$)
- **Singular** ($|A| = 0$)

LINEAR ALGEBRA AND ITS APPLICATIONS

BREAKDOWN OF GAUSSIAN ELIMINATION:

❖ (i) Non-Singular & Curable ($|A| \neq 0$):

❖ Consider the system $x + y + z = 6$

$$x + y + 3z = 10$$

$$x + 2y + 4z = 12$$

$$\begin{pmatrix} 1 & 1 & 1 : 6 \\ 1 & 1 & 3 : 10 \\ 1 & 2 & 4 : 12 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 : 6 \\ 0 & 0 & 2 : 4 \\ 0 & 1 & 3 : 6 \end{pmatrix}$$

Here there is a zero in the second pivot position which can be avoided by a row exchange. Thus breakdown is **temporary** and **curable**.

Hence this system reduces to an upper triangular system which can be solved by Back Substitution and so system will become **consistent** and will have **a unique solution**.

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 : 6 \\ 0 & 1 & 3 : 6 \\ 0 & 0 & 2 : 4 \end{pmatrix} \Rightarrow \left. \begin{array}{l} x + y + z = 6 \\ y + 3z = 6 \\ 2z = 4 \end{array} \right\} \Rightarrow (x, y, z) = (4, 0, 2)$$

LINEAR ALGEBRA AND ITS APPLICATIONS

BREAKDOWN OF GAUSSIAN ELIMINATION:

❖(ii) Singular & InCurable $(|A| = 0)$:

❖ Consider the system

$$\begin{aligned}x + y + z &= 6 \\x + y + 3z &= 10 \\x + y + 4z &= 13\end{aligned}$$

$$\begin{pmatrix} 1 & 1 & 1 : 6 \\ 1 & 1 & 3 : 10 \\ 1 & 1 & 4 : 13 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 : 6 \\ 0 & 0 & 2 : 4 \\ 0 & 0 & 3 : 7 \end{pmatrix}$$

Here there is a zero in the second pivot position which cannot be avoided by any row exchange.

Hence breakdown cannot be avoided which is **incurable**.

The system is **singular** and has **no solution**.

Here we get $z=2$ and $z=7/3$ which is not possible.

$$\Rightarrow \left. \begin{aligned}x + y + z &= 6 \\2z &= 4 \\3z &= 7\end{aligned} \right\}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

BREAKDOWN OF GAUSSIAN ELIMINATION:

❖(iii) Singular ($|A| = 0$):

❖ Consider the system

$$\begin{aligned}x + y + z &= 6 \\x + y + 3z &= 10 \\x + y + 4z &= 12\end{aligned}$$

$$\begin{pmatrix} 1 & 1 & 1 : 6 \\ 1 & 1 & 3 : 10 \\ 1 & 1 & 4 : 12 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 : 6 \\ 0 & 0 & 2 : 4 \\ 0 & 0 & 3 : 6 \end{pmatrix}$$

Here there is a zero in the second pivot position which cannot be avoided. But since last two equations are **consistent** and we get **infinite number of solutions**.

$$\Rightarrow \left. \begin{aligned} x + y + z &= 6 \\ 2z &= 4 \\ 3z &= 6 \end{aligned} \right\} \Rightarrow \begin{aligned} z &= 2 \text{ and } x + y = 4 \\ y &= k \Rightarrow x = 4 - k \end{aligned}$$
$$\Rightarrow (x, y, z) = (4 - k, k, 2)$$

LINEAR ALGEBRA AND ITS APPLICATIONS

BREAKDOWN OF GAUSSIAN ELIMINATION:

(1) Apply Gaussian elimination to the system of equations $u + v + w = -2$

$$3u + 3v - w = 6$$

$$u - v + w = -1$$

When does elimination fail at which pivot position.

Is the breakdown temporary or permanent, discuss.

What coefficient of v in the third equation, in place of the present -1, would make it impossible to proceed-and force elimination to break down?

$$\begin{pmatrix} 1 & 1 & 1 & : & -2 \\ 3 & 3 & -1 & : & 6 \\ 1 & -1 & 1 & : & -1 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 1 & : & -2 \\ 0 & 0 & -4 & : & 12 \\ 0 & -2 & 0 & : & 1 \end{pmatrix}$$

Elimination fails at second pivot position. This breakdown is **temporary** which can be cured by exchanging 2nd and 3rd row. Then system becomes consistent and we get a unique solution.

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & : & -2 \\ 0 & -2 & 0 & : & 1 \\ 0 & 0 & -4 & : & 12 \end{pmatrix} \Rightarrow \left. \begin{array}{l} u + v + w = -2 \\ -2v = 1 \\ -4w = 12 \end{array} \right\}$$

$\Rightarrow (u, v, w) = \left(\frac{3}{2}, -\frac{1}{2}, -3 \right)$ If the coefficient of v in the third equation is 1 instead of -1, then elimination breaks down **permanently** and it is impossible to proceed.

BREAKDOWN OF GAUSSIAN ELIMINATION:

(2) For which three numbers a will elimination fail to give three pivots?

$$ax + 2y + 3z = b_1$$

$$ax + ay + 4z = b_2$$

$$ax + ay + az = b_3$$

$$\begin{pmatrix} a & 2 & 3 : b_1 \\ a & a & 4 : b_2 \\ a & a & a : b_3 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} a & 2 & 3 : b_1 \\ 0 & a-2 & 1 : b_2 - b_1 \\ 0 & a-2 & a-3 : b_3 - b_1 \end{pmatrix}$$
$$\xrightarrow{R_3 - R_2} \begin{pmatrix} a & 2 & 3 : b_1 \\ 0 & a-2 & 1 : b_2 - b_1 \\ 0 & 0 & a-4 : b_3 - b_2 \end{pmatrix}$$

If $a=0$, $a=2$, $a=4$ will fail to give 3 pivots.

LINEAR ALGEBRA AND ITS APPLICATIONS

BREAKDOWN OF GAUSSIAN ELIMINATION:



3. For what values of a and b does the following system have (i) a unique solution
(ii) Infinitely many solutions (iii) No solution.

$$x + 2y + 3z = 2$$

$$-x - 2y + az = -2$$

$$2x + by + 6z = 5$$

$$\begin{pmatrix} 1 & 2 & 3 & : & 2 \\ -1 & -2 & a & : & -2 \\ 2 & b & 6 & : & 5 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{pmatrix} 1 & 2 & 3 & : & 2 \\ 0 & 0 & a+3 & : & 0 \\ 0 & b-4 & 0 & : & 1 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & : & 2 \\ 0 & b-4 & 0 & : & 1 \\ 0 & 0 & a+3 & : & 0 \end{pmatrix}$$

- (i) If $a \neq -3$ and $b \neq 4$ then $r(A)=r(A:b)=3=n$ hence system will be **consistent**

and will have **a unique solution**.

- (ii) If $a = -3$ then $r(A)=r(A:b)=2 < n(=3)$ hence system will be **consistent** and will have infinitely **many solutions**.

- (iii) If $a = -3, b = 4$ (or $a \neq -3, b = 4$), then $r(A)=1$ (or 2) and $r(A:b)=2$ (or 3) hence system will be **inconsistent** and will have **no solution**.



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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Elementary Matrices

- ❖ Elementary Matrix E_{ij} is obtained from the Identity Matrix I by using transformation $R_i - l_{ij}R_j$ where l_{ij} is the multiplier

i.e

$$I \longrightarrow E_{ij}$$

Example: E_{32} is obtained as follows:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} = E_{32}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

Example:

Consider the matrix $A = \begin{pmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 - 1/3 R_1]{R_2 - 2/3 R_1} \begin{pmatrix} 3 & 1 & 2 \\ 0 & -11/3 & -7/3 \\ 0 & 5/3 & 1/3 \end{pmatrix}$

$$\xrightarrow{R_3 + (5/11)R_2} \begin{pmatrix} 3 & 1 & 2 \\ 0 & -11/3 & -7/3 \\ 0 & 0 & -8/11 \end{pmatrix} = U \quad E = E_{21} = R_2 - (2/3)R_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2/3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$F = E_{31} = R_3 - (1/3)R_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/3 & 0 & 1 \end{pmatrix} \quad G = E_{32} = R_3 + (5/11)R_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5/11 & 1 \end{pmatrix}$$

$$E_{32}E_{31}E_{21}A = U$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

Problems:

1. Which elimination matrices put A into upper triangular matrix U?

$$\begin{aligned}
 A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} &\xrightarrow{R_2 + (1/2)R_1} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \xrightarrow{R_3 + (2/3)R_2} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \\
 &\xrightarrow{R_4 + (3/4)R_3} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & 0 & 5/4 \end{pmatrix} = U
 \end{aligned}$$

The elimination matrices E_{21} , E_{32} and E_{43} put A into upper triangular matrix i.e.,

$$E_{43}E_{32}E_{21}A = U$$

$$E_{43}E_{32}E_{21}A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 3/4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2/3 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} A = U$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

2. Which elementary matrices convert $A = \begin{pmatrix} 0 & 2 & -6 & -2 & 4 \\ 0 & -1 & 3 & 3 & 2 \\ 0 & -1 & 3 & 7 & 10 \end{pmatrix}$ into upper triangular matrix U?

$$A = \begin{pmatrix} 0 & 2 & -6 & -2 & 4 \\ 0 & -1 & 3 & 3 & 2 \\ 0 & -1 & 3 & 7 & 10 \end{pmatrix} \xrightarrow[R_3 + 1/2 R_1]{R_2 + 1/2 R_1} \begin{pmatrix} 0 & 2 & -6 & -2 & 4 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 & 12 \end{pmatrix}$$

$$\xrightarrow{R_3 - 3R_2} \begin{pmatrix} 0 & 2 & -6 & -2 & 4 \\ 0 & 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = U$$

Matrices E_{21}, E_{31}, E_{32} convert
A into triangular form U.

$$E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{pmatrix} \quad E_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix}$$



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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Triangular Factors

- ❖ Consider the system of equations $Ax=b$. Using elementary row transformations we reduce A to U an Upper triangular matrix. While doing so $Ax=b$ reduces to $Ux=c$ (i.e. $[b]$ reduces to $[c]$) which can be solved by back substitution to obtain x . The steps involved are $Ax = b \Rightarrow E_{32}E_{31}E_{21}Ax = Ux = c \left(\because E_{32}E_{31}E_{21}A = U \right)$
- To undo the steps of elimination we must trace back the steps instead of subtracting we must add. For this we need inverses of Elementary Matrices i.e. $E_{21}^{-1}, E_{31}^{-1}, E_{32}^{-1}$
- E_{21}^{-1} can be obtained by changing -1 to 1 in the transformation. Similarly E_{31}^{-1} and E_{32}^{-1} can also be obtained. But while going from U to A the inverses should be in reverse direction i.e., $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U = A$

Triangular Factorization $A=LU$:

- ❖ Any square matrix A can be factored as $A=LU$ where
 - L is a Lower Triangular Matrix
 - With 1's on the diagonal
 - Having multiplier's l_{ij} below the diagonal in their respective positions.
- And
 - U is an Upper Triangular Matrix
 - Having pivots on the diagonal (u_{11}, u_{22}, u_{33})

$$\text{➤ i.e } L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix} = \begin{pmatrix} d_1 & u_{12} & u_{13} \\ 0 & d_2 & u_{23} \\ 0 & 0 & d_3 \end{pmatrix}$$

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MATRICES AND GAUSSIAN ELIMINATION:

Consider the matrix $A = \begin{pmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{pmatrix} \xrightarrow[R_3 - 1/3 R_1]{R_2 - 2/3 R_1} \begin{pmatrix} 3 & 1 & 2 \\ 0 & -11/3 & -7/3 \\ 0 & 5/3 & 1/3 \end{pmatrix}$

$$\xrightarrow{R_3 + (5/11)R_2} \begin{pmatrix} 3 & 1 & 2 \\ 0 & -11/3 & -7/3 \\ 0 & 0 & -8/11 \end{pmatrix} = U \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & -5/11 & 1 \end{pmatrix}$$

$$A = LU = \begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & -5/11 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 \\ 0 & -11/3 & -7/3 \\ 0 & 0 & -8/11 \end{pmatrix}$$

Triangular Factorization $A=LDU$:

- ❖ $A=LU$ factorization is Unsymmetric as L has 1's on the diagonal whereas U has pivots on the diagonal. In order to make this factorization symmetric we do $A=LDU$ factorization
- **D** is a Diagonal Matrix with pivots d_1, d_2, d_3 on the diagonal
- **L** is a Lower Triangular Matrix
- With 1's on the diagonal
- Having multiplier's l_{ij} below the diagonal in their respective positions.
- **U** is an Upper Triangular Matrix with 1's on the diagonal obtained by dividing each row by its pivot.

i.e

$$L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \quad D = \begin{pmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{pmatrix} \quad U = \begin{pmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} \\ 0 & 1 & u_{23}/u_{22} \\ 0 & 0 & 1 \end{pmatrix}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

Factorise $A=LU$ and hence $A=LDU$.

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \xrightarrow{R_2 + (1/2)R_1} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \xrightarrow{R_3 + (2/3)R_2} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_4 + (3/4)R_3} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & 0 & 5/4 \end{pmatrix} = U \quad A = LU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 0 & -2/3 & 1 & 0 \\ 0 & 0 & -3/4 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 3/2 & -1 & 0 \\ 0 & 0 & 4/3 & -1 \\ 0 & 0 & 0 & 5/4 \end{pmatrix}$$

$$A = LDU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 0 & -2/3 & 1 & 0 \\ 0 & 0 & -3/4 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3/2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \\ 0 & 0 & 0 & 5/4 \end{pmatrix} \begin{pmatrix} 1 & -1/2 & 0 & 0 \\ 0 & 1 & -2/3 & 0 \\ 0 & 0 & 1 & -3/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

❖ To solve a system of equations $Ax=b$, factorize $A=LU$, then $Ax=LUx$ where

Let $Ux=c$, then $Lc=b=Ax$.

Solve $Lc=b$, using Forward elimination and then find x using $Ux=c$ by Backward substitution which gives x .

Ex: Solve as triangular systems without multiplying LU to find A.

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \quad \text{Let } Lc = b \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$
$$\Rightarrow \begin{cases} c_1 = 2 \\ c_2 = -2 \\ c_3 = 0 \end{cases} \text{ then } Ux = c \Rightarrow \begin{pmatrix} 2 & 4 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2x + 4y + 4z = 2 \\ y + 2z = -2 \\ z = 0 \end{cases}$$
$$(x, y, z) = (5, -2, 0)$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

❖ Find $A=LU$ and LDU factorization given

$$\begin{aligned}
 A &= \begin{pmatrix} 6 & -2 & -4 & 4 \\ 3 & -3 & -6 & 1 \\ -12 & 8 & 21 & -8 \\ -6 & 0 & -10 & 7 \end{pmatrix} \xrightarrow[\begin{smallmatrix} R_3+2R_1 \\ R_4+R_1 \end{smallmatrix}]{R_2-1/2R_1} \begin{pmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 4 & 13 & 0 \\ 0 & -2 & -14 & 11 \end{pmatrix} \xrightarrow[\begin{smallmatrix} R_4-R_2 \end{smallmatrix}]{R_3+2R_2} \begin{pmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & -10 & 12 \end{pmatrix} \\
 &\xrightarrow{R_4+2R_3} \begin{pmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & 0 & 8 \end{pmatrix} = U \quad A = LU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ -2 & -2 & 1 & 0 \\ -1 & 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 6 & -2 & -4 & 4 \\ 0 & -2 & -4 & -1 \\ 0 & 0 & 5 & -2 \\ 0 & 0 & 0 & 8 \end{pmatrix} \\
 A = LDU &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ -2 & -2 & 1 & 0 \\ -1 & 1 & 5/2 & 1 \end{pmatrix} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix} \begin{pmatrix} 1 & -1/3 & -2/3 & 2/3 \\ 0 & 1 & 2 & 1/2 \\ 0 & 0 & 1 & -2/5 \\ 0 & 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$



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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Row Exchanges and Permutation Matrices

❖ Consider the system of equations $Ax=b$. While solving the system if a zero appears in the pivot position, it calls for a row exchange.

This row exchange is taken care by **Permutation Matrices P**.

Here $A \neq LU$ then $PA=LU$ where **P is a Permutation Matrix** which is an Identity Matrix with rows in different order.

Ex: Consider the system $y=b_1; 2x-3y=b_2$

$Ax = b \Rightarrow \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ Here Gaussian elimination fails and so calls for a row exchange i.e., $\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$

This is same as $PAX=Pb$

$$PA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} = U \text{ and } Pb = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$$

$$\therefore PAX = PB \Rightarrow \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$$

Permutation Matrices:

- ❖ **P is a Permutation Matrix** which is an Identity Matrix with rows in different order.
- **Product of two permutation matrices is also a Permutation Matrix.**
- **Inverse of a permutation matrices is also a Permutation Matrix.**
- **P^{-1} is always same as P^T .**
- **Permutation Matrices of order 2 are $2!=2$ in number.**

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad P_{21} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

❖ Permutation Matrices of order 3 are $3!=6$ in number.

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} ; \quad P_{21} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = P_{21}^{-1} ;$$

$$P_{31} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = P_{31}^{-1} ; \quad P_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = P_{32}^{-1} ;$$

$$P_{21}P_{31} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = (P_{21}P_{32})^{-1} ; \quad P_{21}P_{32} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = (P_{21}P_{32})^{-1}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

$$\text{Ex: } A = \begin{pmatrix} 2 & 3 & 3 \\ 6 & 9 & 8 \\ 0 & 5 & 7 \end{pmatrix} \xrightarrow{R_2 - 3R_1} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 0 & -1 \\ 0 & 5 & 7 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = U$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} \neq A$$

$$P_{23}A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 6 & 9 & 8 \\ 0 & 5 & 7 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 6 & 9 & 8 \end{pmatrix} \xrightarrow{R_3 - 3R_1} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = U$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = P_{23}A$$

Problem: Explain why A is not factorizable into LU? How can A be modified so that the new matrix can be factored into LU? Also obtain the factors L,D,U for the new matrix. What is the relation between L and U thus obtained? Explain.

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -4 & 5 \\ -2 & 5 & -4 \end{pmatrix} \xrightarrow[R_3+2R_1]{R_2-2R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A is not factorizable into LU as Gaussian elimination fails and Row exchange is required.

So A should be multiplied with permutation matrix P_{23} so that $PA=LU$.

$$P_{23}A = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 5 & -4 \\ 2 & -4 & 5 \end{pmatrix} \xrightarrow[R_3-2R_1]{R_2+2R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = U \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad L=U^T$$



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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Inverses and Transposes

- ❖ Let A be a square matrix of order n the **Inverse of A** is the matrix B such that **$AB=I=BA$** .
Here $B=A^{-1}$.
- **Properties:** Inverse of a matrix is unique. i.e., $AB=BA=I$ and $AC=CA=I$, then $B=C$
- Inverse of the product is the product of Inverses. $(ABCD)^{-1}=D^{-1} C^{-1} B^{-1} A^{-1}$
- If $A=LU$ then $A^{-1}=U^{-1} L^{-1}$.
- Since $E_{32}E_{31}E_{21}A = U$ we have $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U = A \Rightarrow A^{-1} = U^{-1}E_{32}E_{31}E_{21}$
$$\Rightarrow L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$$
- A matrix A is invertible if and only if elimination produces n pivots with or without row exchanges. Elimination solves $Ax=b$ without explicitly finding A^{-1} .
- If A is invertible, the one and only one solution to $Ax=b$ is $x= A^{-1} b$

Gauss Jordan Method of computing A^{-1} :

- ❖ The inverse of an invertible matrix is obtained by a set of row operations that transforms A to I and I to A^{-1} . This process is known as **Gauss Jordan Method**.
- ❖ Consider the Augmented Matrix **$[A:I]$** . Then perform row operations on it so that A reduces to Echelon form **U** and at the same time **I** reduces to **C** . Further reduce **U to I** using elementary row transformations which reduces **C to A^{-1}** .

❖ i.e., $[A:I] \rightarrow [U:C] \rightarrow [I:A^{-1}]$

Ex: Compute A^{-1} using Gauss Jordan Method given

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 4 \\ -2 & 2 & 2 \end{pmatrix}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

$$\begin{aligned} \text{Ex: } [A:I] &= \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 4 & 3 & 4:0 & 1 & 0 \\ -2 & 2 & 2:0 & 0 & 1 \end{pmatrix} \xrightarrow[R_3 + R_1]{R_2 - 2R_1} \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 0 & 1 & 2:-2 & 1 & 0 \\ 0 & 3 & 3:1 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{R_3 - 3R_2} \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 0 & 1 & 2:-2 & 1 & 0 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} = [U:C] \xrightarrow[R_2 + 2/3R_3]{R_1 + 1/3R_3} \begin{pmatrix} 2 & 1 & 0:10/3 & -1 & 1/3 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} \\ &\xrightarrow{R_1 - R_2} \begin{pmatrix} 2 & 0 & 0:2/3 & 0 & -1/3 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} \xrightarrow[R_3 = -1/3R_3]{R_1 = 1/2R_1} \begin{pmatrix} 1 & 0 & 0:1/3 & 0 & -1/6 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & 1:-7/3 & 1 & -1/3 \end{pmatrix} \\ &= [I:B] \\ B = A^{-1} &= \begin{pmatrix} 1/3 & 0 & -1/6 \\ 8/3 & -1 & 2/3 \\ -7/3 & 1 & -1/3 \end{pmatrix} \end{aligned}$$

Transpose of a Matrix A^T :

❖ If $A = [a_{ij}]_{m \times n}$ is an $m \times n$ matrix, then its transpose is obtained by interchanging its rows and columns and is denoted by $A^T = [a_{ji}]_{n \times m}$.

Ex: If $A = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \end{pmatrix}_{2 \times 3}$ then $A^T = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ -3 & 0 \end{pmatrix}_{3 \times 2}$

➤ Properties:

➤ The Transpose of a Lower Triangular Matrix is an Upper Triangular Matrix.

$$(A^T)^T = A ; (AB)^T = B^T A^T ; (A^{-1})^T = (A^T)^{-1} ;$$

$$(A \pm B)^T = A^T \pm B^T ; (A^{-1})^T A^T = (AA^{-1})^T = I$$

Symmetric Matrices:

❖ If A is a matrix of order n is said to be symmetric matrix of order if $A^T = A$

Ex: If $A = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$ then $A^T = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$

➤ Properties:

- If A is symmetric then A^{-1} may or may not exist.
- If for a symmetric matrix A^{-1} exists then A^{-1} is also symmetric.
- For a symmetric matrix A , we have $(A^{-1})^T = (A)^{-1}$ ($\because A^T = A$)

Symmetric Products AA^T , $A^T A$, LDL^T :

❖ If A is a matrix of order $m \times n$, then AA^T and $A^T A$ are both symmetric

Ex: If $A = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \end{pmatrix}_{2 \times 3}$ then $AA^T = \begin{pmatrix} 14 & 10 \\ 10 & 20 \end{pmatrix}$ $A^T A = \begin{pmatrix} 20 & 10 & -6 \\ 10 & 5 & -3 \\ -6 & -3 & 9 \end{pmatrix}$

➤ If A is symmetric and if $A = LDU$ then,

$$A = A^T = LDL^T \quad (\because U = L^T \text{ \& } L = U^T)$$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{pmatrix} \xrightarrow[R_3 + R_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 2 & 3 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 7 \end{pmatrix} = U$$

$$LDU = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \quad U = L^T$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

❖ For which three numbers “c” is this matrix not invertible, and why not?

$$A = \begin{pmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{pmatrix} \xrightarrow[R_3 - 4R_1]{R_2 - c/2R_1} \begin{pmatrix} 2 & c & c \\ 0 & c - (c^2/2) & c - (c^2/2) \\ 0 & 7 - 4c & -3c \end{pmatrix}$$
$$\xrightarrow{R_3 - \left(\frac{7 - 4c}{2c - c^2}\right)2R_2} \begin{pmatrix} 2 & c & c \\ 0 & c - (c^2/2) & c - (c^2/2) \\ 0 & 0 & c - 7 \end{pmatrix}$$

- Matrix A is not invertible for $c=0,2,7$
- For $c=0,2,7$ elimination gives one zero row, hence A will be singular and so A will not be invertible.



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LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Supplementary problems

1. Find PA=LU and PA=LDU for $A = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

$$A = \begin{pmatrix} 3 & -1 & 0 \\ 6 & -2 & 0 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow[R_3 - 1/3 R_1]{R_2 - 2R_1} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 4/3 & 2 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$PA = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 1 & 2 \\ 6 & -2 & 0 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - 1/3 R_1} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix} = U$$

$$PA = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & 0 \\ 0 & 4/3 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4/3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{pmatrix}$$

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MATRICES AND GAUSSIAN ELIMINATION:



2. For what values of a and b does the following system have (i) a trivial solution

(ii) Infinitely many solutions.

$$\begin{aligned}x + 2y + 3z &= 0 \\ -x - 2y + az &= 0 \\ 2x + by + 6z &= 0\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & a \\ 2 & b & 6 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & a+3 \\ 0 & b-4 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & b-4 & 0 \\ 0 & 0 & a+3 \end{pmatrix}$$

This system will have only either a trivial solution or infinitely many solutions. It will have

(i) **a trivial solution** if $a \neq -3$ & $b \neq 4$ then $r(A)=3=n$

(ii) **Infinitely many solutions** if $a = -3$ or $b = 4$ or *both* then $r(A)$ will be 2 or 1 respectively.

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GAUSSIAN ELIMINATION:



3. Check for consistency and solve the following system of equations if consistent. Also discuss its rank:

$$\begin{array}{l} 3x + y + 2z = 3 \\ 2x - 3y - z = -3 \\ x + 2y + z = 4 \end{array} \quad \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 2 & -3 & -1 & -3 \\ 1 & 2 & 1 & 4 \end{array} \right) \xrightarrow{\begin{array}{l} R_2 - \left(\frac{2}{3}\right)R_1 \\ R_3 - \left(\frac{1}{3}\right)R_1 \end{array}} \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 0 & -11/3 & -7/3 & -5 \\ 0 & 5/3 & -1/3 & 3 \end{array} \right)$$

$$\xrightarrow{R_3 + \left(\frac{5}{11}\right)R_2} \left(\begin{array}{ccc|c} 3 & 1 & 2 & 3 \\ 0 & -11/3 & -7/3 & -5 \\ 0 & 0 & -24/33 & 8/11 \end{array} \right) \Rightarrow \begin{cases} 3x + y + 2z = 3 \\ (-11/3)y - 7/3z = -5 \\ (-24/33)z = 8/11 \end{cases}$$

$r(A)=r(A:b)=3=n$ hence system is **consistent** and has **a unique solution**.

i.e **$(x, y, z)=(1, 2, -1)$** . Its rank is 3.

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4. Find an LU and LDU factorization for A. What is the rank of A?

$$\begin{aligned}
 A &= \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{pmatrix} \xrightarrow[\substack{R_3 - R_1 \\ R_4 + 3R_1}]{R_2 + 2R_1} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{pmatrix} \xrightarrow[\substack{R_4 - 4R_2}]{R_3 + 3R_2} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{pmatrix} \\
 &\xrightarrow{R_4 - 2R_3} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \quad A = LU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix} \\
 A = LDU &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1/2 & 5/2 & -1 \\ 0 & 1 & 1/3 & 2/3 & -1 \\ 0 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

Rank of A is 4.

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5. Solve $Ax=b$ for x if $A=\begin{pmatrix} 1 & 0 & -2 \\ 2 & 1 & 3 \\ 4 & 2 & 5 \end{pmatrix}$ and $b=\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$

$$[A:I]=\begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 2 & 1 & 3 & : & 0 & 1 & 0 \\ 4 & 2 & 5 & : & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_3-4R_1]{R_2-2R_1} \begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 0 & 1 & 7 & : & -2 & 1 & 0 \\ 0 & 2 & 13 & : & -4 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{R_3-2R_2} \begin{pmatrix} 1 & 0 & -2 & : & 1 & 0 & 0 \\ 0 & 1 & 7 & : & -2 & 1 & 0 \\ 0 & 0 & -1 & : & 0 & -2 & 1 \end{pmatrix} \xrightarrow[R_2+7R_3]{R_1-2R_3} \begin{pmatrix} 1 & 0 & 0 & : & 1 & 4 & -2 \\ 0 & 1 & 0 & : & -2 & -13 & 7 \\ 0 & 0 & -1 & : & 0 & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{-R_3} \begin{pmatrix} 1 & 0 & 0 & : & 1 & 4 & -2 \\ 0 & 1 & 0 & : & -2 & -13 & 7 \\ 0 & 0 & 1 & : & 0 & 2 & -1 \end{pmatrix} \quad A^{-1}=\begin{pmatrix} 1 & 4 & -2 \\ -2 & -13 & 7 \\ 0 & 2 & -1 \end{pmatrix} \quad x=\begin{pmatrix} 0 \\ 4 \\ -1 \end{pmatrix}$$



THANK YOU

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